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Affine FR: an Effective Facial Reduction Algorithm for Semidefinite Relaxations of Combinatorial Problems

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Abstract We propose a novel method, *affine facial reduction (affine FR)*, for restoring Slater's condition in semidefinite programming (SDP) relaxations of combinatorial optimization (CO) problems. We provide a rigorous comparison between affine FR and existing facial reduction methods. Numerical experiments further demonstrate the effectiveness of affine FR in reducing the size of SDP relaxations for CO problems.

Keywords facial reduction · exposing vector · semidefinite programming · Slater's condition · combinatorial optimization

Mathematics Subject Classification (2020) 65K05 · 90C27

1 Introduction

We present a novel facial reduction algorithm (FRA) tailored for semidefinite programming (SDP) relaxations of combinatorial optimization (CO) problems. The algorithm is specifically designed for CO problems involving binary variables, which naturally represent yes/no decisions in various applications. Such problems commonly arise in areas including radio frequency assignment, timetabling, scheduling, cargo transportation, large network communication, satellite network design, and resource allocation.

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CO problems are mathematically challenging, and state-of-the-art solution methods often struggle to efficiently solve large-scale instances that are crucial in both business and scientific applications. In contrast, SDP is a class of convex optimization problems characterized by a linear objective function and affine constraints over the cone of positive semidefinite matrices, see [2, 45]. SDP has attracted significant attention in the CO community over the past three decades due to its ability to provide tight bounds on the optimal value, thereby accelerating the solution process. Notable success has been achieved using SDP relaxations for challenging CO problems such as the max-cut problem [15,36] and the quadratic assignment problem [47]. A specialized first-order method for optimizing lift-and-project relaxations of binary integer programs is proposed in [8]. Despite recent advances in first-order methods that have improved the practicality of SDP-based approaches, significant challenges remain — particularly in handling large-scale problems and managing numerical instability caused by the lack of regularity conditions in many SDPs [13,22,30,48]. More precisely, the dimension of SDP relaxations grows rapidly with problem size, and it is well known that SDP solvers do not scale as effectively as linear programming (LP) solvers. Due to the inherent structure of many CO problems, the corresponding SDP relaxations often fail to satisfy regularity conditions, such as Slater's condition, leading to numerical instability during optimization.

Facial reduction is a preprocessing technique introduced by Borwein and Wolkowicz to regularize SDPs by restoring Slater's condition [6,7]. It has proven effective in various contexts, including the quadratic assignment problem [47], where restoring Slater's condition can significantly accelerate computation. Although facial reduction is theoretically guaranteed to succeed, implementing it in practice is highly nontrivial, prompting the development of specialized methods [33,49].

In this paper, we introduce *affine FR*, a new implementation of the FRA, tailored for SDP relaxations of CO problems. Our approach leverages the geometric structure inherent in the CO problems and its interplay with their SDP relaxations. Through theoretical analysis and numerical experiments, we demonstrate several key advantages of affine FR:

- Effectiveness: By exploiting the problem structure, affine FR outperforms existing general-purpose approaches for CO problems.
- Applicability: Affine FR applies to a broader class of CO problems than existing facial reduction techniques.
- User-friendliness: Affine FR is fully automatic and requires only a description of the problem, minimizing the user's involvement in preprocessing.

Affine FR is broadly applicable to mixed-binary integer programming problems and reduces the effort required in the preprocessing stage. This contributes to making SDPs a more practical and scalable tool for operations research and beyond.

Notation. We use $\mathbb{S}^n$, $\mathbb{S}^n_+$ and $\mathbb{S}^n_{++}$ to denote the sets of $n \times n$ real symmetric matrices, positive semidefinite matrices, and positive definite matrices,

respectively. For two matrices of the same size, the notation $\langle \cdot, \cdot \rangle$ refers to their Frobenious inner product. For a vector $u = (u_1, \ldots, u_n)$, we denote its componentwise absolute value by $|u| = (|u_1|, \ldots, |u_n|)$. The standard basis vectors in $\mathbb{R}^n$ are denoted by $e_1, \ldots, e_n$. When lifting a set from $\mathbb{R}^n$ to $\mathbb{R}^{n+1}$, we occasionally abuse notation and refer to the standard basis vectors in $\mathbb{R}^{n+1}$ as $e_0, \ldots, e_n$. We denote the all-ones and all-zeros vectors of length k by $\mathbf{1}_k$ and $\mathbf{0}_k$, respectively. The $k \times l$ matrix of all-zeros is denoted by $\mathbf{O}_{k\times l}$; when $k = l$, we write $\mathbf{O}_k$. The $k \times k$ identity matrix is denoted by I_k. When the dimensions are clear from the context, we omit the subscripts and simply write $\mathbf{1}, \mathbf{0}, \mathbf{O}$ and I. We use $\mathrm{Diag}(u)$ to denote the diagonal matrix whose diagonal entries are given by the components of u. For a subset S of $\mathbb{R}^n$ or $\mathbb{S}^n$, we use $\mathrm{int}(S)$, $\mathrm{relint}(S)$, and $S^\perp$ to denote its interior, relative interior, and the orthogonal complement of the span of S, respectively. The convex hull, affine hull, dimension, and cardinality of S are denoted by $conv(S)$, $\mathrm{aff}\,(S)$, $\dim(S)$, and $|S|$, respectively. For a closed convex cone $\mathcal{K}$, we write $\mathcal{K}^*$ for its dual cone. Given a linear operator $\mathcal{A}$, we use $\mathcal{A}^*$ to denote its adjoint. The intended meaning of the star notation will be clear from context.

2 Preliminaries

2.1 SDP relaxation for CO problems

CO problems aim to find optimal solutions involving discrete variables. To simplify the presentation, we focus on CO problems whose feasible region F is the intersection of the binary hypercube and a polyhedron P, that is,

$$F = P \cap \{0, 1\}^n. \tag{1}$$

We assume that the polyhedron P is defined by a linear system $Ax \leq b$, that is, $P = \{x \in \mathbb{R}^n \mid Ax \leq b\}$. Although we focus here on purely binary problems, our method also applies to more general mixed-binary linear programming problems which contain both continuous and binary variables; see Section 5 and Remark 2. Throughout, we assume that P is nonempty and that each binary variable x_i is between zero and one for any $x \in P$.

Assumption 1 *$P \neq \emptyset$. If $x \in P$, then $\boldsymbol{0} \leq x \leq \boldsymbol{1}$.*

Our goal is to maximize or minimize a given objective function over the set F. In many applications, the objective function is linear or quadratic, in which case SDP relaxations are commonly used, as they often yield tight bounds on the optimal value. SDP is a class of convex optimization problems with linear objective functions and affine constraints over the cone of positive semidefinite matrices. To construct an SDP relaxation for the feasible set F, we consider a lifted feasible set

$$F_1 := \left\{ \begin{bmatrix} 1 \\ x \end{bmatrix} \begin{bmatrix} 1 \\ x \end{bmatrix}^T \,\middle|\, x \in F \right\} \subseteq \mathbb{S}^{n+1}_+. \tag{2}$$

We introduce a matrix variable $Y \in \mathbb{S}^{n+1}$ to relax the nonlinear expression $\left[\begin{smallmatrix}1\\x\end{smallmatrix}\right]\left[\begin{smallmatrix}1\\x\end{smallmatrix}\right]^T$ in F_1. Clearly, the matrix variable Y must be positive semidefinite. Additional valid linear equality and inequality constraints on Y can be added to strengthen the relaxation. In general, let $L \subseteq \mathbb{S}^{n+1}$ be a polyhedron such that $F_1 \subseteq L$, then $F_1 \subseteq L \cap \mathbb{S}^{n+1}_+$ and the intersection

$$L \cap \mathbb{S}^{n+1}_+$$

defines an SDP relaxation for F_1. Since there is a one-to-one correspondence between elements of F_1 and those of F, we also refer to $L \cap \mathbb{S}^{n+1}_+$ as an SDP relaxation of $F \subseteq \mathbb{R}^n$ for convenience.

In practice, many SDP relaxations exist, each offering different tradeoffs between tightness and computational efficiency. One of the most widely used is *Shor's SDP relaxation*, which we describe in detail in (3). The design of a good SDP relaxation involves balancing its accuracy against the computational effort required to solve it. For a broad discussion on the construction of SDP relaxations for CO problems, see [35,44].

Shor's relaxation is particularly relevant for *quadratically constrained quadratic programs (QCQPs)*, of which the mixed-binary linear programs considered in this paper form a special case. A comprehensive comparison of various SDP relaxations for QCQP is presented in [4], and the exactness of these relaxations, including Shor's relaxation, is studied in [43].

While SDP relaxations are known to provide strong bounds for CO problems, they come with practical challenges. In particular, SDPs scale poorly with problem size and can be computationally expensive. Moreover, due to the specific structure of CO problems, SDP relaxations often exhibit degeneracy. A theoretical framework for addressing these issues is discussed in Section 2.3.

2.2 Shor's SDP relaxation

The proposed method, affine FR, can be applied to regularize any SDP relaxation; see Section 3.2. Our theoretical analysis in Section 4 focuses on Shor's SDP relaxation (3), which we formally define below. Let F be the binary feasible set defined in (1). Shor's SDP relaxation of F is given by $L \cap \mathbb{S}^{n+1}_+$, where L is defined as follows:

$$L := \left\{ Y = \begin{bmatrix} Y_{00} & Y_{01} & \cdots & Y_{0n} \\ Y_{10} & Y_{11} & \cdots & Y_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ Y_{n0} & Y_{n1} & \cdots & Y_{nn} \end{bmatrix} \in \mathbb{S}^{n+1} \;\middle|\; Arrow(Y) = e_0,\ \mathcal{A}(Y) \leq \mathbf{0} \right\}. \quad (3)$$

Here, $e_0 = (1, 0, \ldots, 0)^T \in \mathbb{R}^{n+1}$ is the first standard basis vector. The operator $\mathcal{A} : \mathbb{S}^{n+1} \to \mathbb{R}^m$ encodes the linear constraints of the original problem and is

defined by

$$(\mathcal{A}(Y))_i := \left\langle \begin{bmatrix} -b_i & \frac{1}{2}a_i^T \\ \frac{1}{2}a_i & \mathbf{O} \end{bmatrix}, Y \right\rangle \text{ for } i = 1, \ldots, m, \tag{4}$$

where $a_i^T x \leq b_i$ is the i-th constraint in the system $Ax \leq b$. The arrow operator $Arrow : \mathbb{S}^{n+1} \to \mathbb{R}^{n+1}$ is given by

$$Arrow(Y) := \begin{bmatrix} Y_{00} \\ Y_{11} - \frac{1}{2}(Y_{01} + Y_{10}) \\ \vdots \\ Y_{nn} - \frac{1}{2}(Y_{0n} + Y_{n0}) \end{bmatrix} \in \mathbb{R}^{n+1}. \tag{5}$$

The arrow constraint enforces $Y_{00} = 1$ and $Y_{0i} = Y_{ii}$ for all $i = 1, \ldots, n$, serving as a relaxation of the nonlinear binary condition $x_i = x_i^2$ from (1). Accordingly, the matrix variable Y in this relaxation takes the form

$$Y = \begin{bmatrix} 1 & x^T \\ x & X \end{bmatrix} \text{ for some } x \in \mathbb{R}^n, X \in \mathbb{S}^n \text{ such that } x_i = X_{ii} \text{ for } i = 1, \ldots, n.$$

2.3 Facial reduction

We say that *Slater's condition* holds for the set $L \cap \mathbb{S}^n_+$ if $L \cap \operatorname{int}(\mathbb{S}^n_+) \neq \emptyset$, i.e., there exists a positive definite feasible solution $Y \in L \cap \mathbb{S}^n_+$. When Slater's condition is satisfied, optimization over $L \cap \mathbb{S}^n_+$ is numerically stable. In contrast, when Slater's condition fails, solvers may encounter serious numerical issues or fail to find the correct optimal solution. In such cases, we seek an equivalent reformulation of $L \cap \mathbb{S}^n_+$ that does satisfy Slater's condition. If such a reformulation is found, we say that Slater's condition has been restored.

Facial reduction is a theoretical framework for restoring Slater's condition originally developed by Borwein and Wolkowicz in [6,7]. Practical implementations of facial reduction are called *facial reduction algorithms* (FRAs), and various FRAs have been proposed. While the theory of facial reduction is well-established, its practical implementation is still in its infancy.

An essential concept in facial reduction is that of *exposing vectors*, which we now explain:

1. If $W \in \mathbb{S}^n_+$, then the set
$$\{Y \in \mathbb{S}^n_+ \mid \langle W, Y \rangle = 0\} \tag{6}$$
is a face of $\mathbb{S}^n_+$, and W is called an *exposing vector* of this face.[1]
2. A matrix W is an *exposing vector for* $L \cap \mathbb{S}^n_+$, if the face exposed by W contains the feasible region, i.e.,
$$\langle W, Y \rangle = 0 \quad \forall\, Y \in L \cap \mathbb{S}^n_+. \tag{7}$$
In particular, if W has the maximum rank among such matrices, it exposes the smallest face of $\mathbb{S}^n_+$ containing $L \cap \mathbb{S}^n_+$.

[1] While the variables in SDP are matrices, the term "vectors" remains standard terminology.

In [6,7], the authors construct exposing vectors using the hyperplane separation theorem. Specifically, given a polyhedron L and a closed convex cone $\mathcal{K} \subseteq \mathbb{S}^n$, the condition $L \cap \operatorname{relint}(\mathcal{K}) = \emptyset$ holds if and only if there exists a nonzero $W \in \mathbb{S}^n$ such that

$$\langle W, X \rangle \leq \langle W, Y \rangle \text{ for all } X \in L \text{ and } Y \in \operatorname{relint}(\mathcal{K}). \tag{8}$$

This means that the hyperplane defined by W separates L and $\operatorname{relint}(\mathcal{K})$. Denote by

$$SEP(L \cap \mathcal{K})$$

the set of all matrices $W \in \mathbb{S}^n$ that satisfy (8), including the zero matrix.[2] Every nonzero matrix in $SEP(L \cap \mathbb{S}^n_+)$ is an exposing vector for $L \cap \mathbb{S}^n_+$. However, an exposing vector for $L \cap \mathbb{S}^n_+$ is not necessarily in $SEP(L \cap \mathbb{S}^n_+)$. If L is an affine subspace, as in standard-form SDPs, then $SEP(L \cap \mathbb{S}^n_+)$ admits a simple description:

$$SEP(L \cap \mathbb{S}^n_+) = L^\perp \cap \mathbb{S}^n_+. \tag{9}$$

An exposing vector allows us to regularize $L \cap \mathbb{S}^n_+$. For example, if $W = e_1 e_1^T$ is an exposing vector (where e_1 is the first standard basis vector), then equation (7) implies that the first row and column of any feasible $Y \in L \cap \mathbb{S}^n_+$ must be zero. This allows us to reduce the dimension of the matrix variable: we can replace $Y \in \mathbb{S}^n_+$ with a smaller matrix variable $R \in \mathbb{S}^{n-1}_+$ by removing the first row and column, and reformulate the problem accordingly.

More generally, if W is any nonzero exposing vector for $L \cap \mathbb{S}^n_+$, we can reformulate $L \cap \mathbb{S}^n_+$ equivalently as

$$\tilde{L} \cap \mathbb{S}^r_+ \text{ where } \tilde{L} := \left\{ R \in \mathbb{S}^r \mid VRV^T \in L \right\}. \tag{10}$$

Here, r is the dimension of the null space of W, and $V \in \mathbb{R}^{n \times r}$ is a matrix whose columns form a basis for that null space. We refer to V as the *facial range vector* associated with the face exposed by W.

The facial reduction theory shows that $L \cap \mathbb{S}^n_+$ satisfies Slater's condition if and only if $SEP(L \cap \mathbb{S}^n_+) = \{\mathbf{O}\}$. If a nonzero $W \in SEP(L \cap \mathbb{S}^n_+)$ exists, we can use its associated facial range vector V to reduce the problem size via (10). The process of identifying such a W and reformulating the problem is called an *FR step*. Repeating this step until no further exposing vectors can be found yields a *complete facial reduction,* which guarantees that the final problem satisfies Slater's condition.

Assuming that the feasibility problems over $SEP(L \cap \mathbb{S}^n_+) \setminus \{\mathbf{O}\}$ can be solved, Slater's condition can always be restored in at most n steps. At each step, choosing a maximum-rank exposing vector minimizes the number of steps required. The minimum number of FR steps needed for an FRA to restore Slater's condition is known as the *singularity degree* of $L \cap \mathbb{S}^n_+$ [39]. This quantity is also crucial for deriving error bounds for linear matrix inequalities and plays an important role in conic optimization [11,20,24,27,31,32,38,40].

[2] In this section, we focus on the case where $\mathcal{K} = \mathbb{S}^n_+$. However, we define $SEP(L \cap \mathcal{K})$ for a general closed convex cone $\mathcal{K}$, as this more general form will be used in the analyses in Sections 2.5 and 4.2.

Note that the singularity degree of $L \cap \mathbb{S}^n_+$ is not always one. Even a maximum-rank element of $SEP(L \cap \mathbb{S}^n_+)$ may not expose the smallest face containing $L \cap \mathbb{S}^n_+$, as this set may not contain all exposing vectors. Hence, a single FR step may not restore Slater's condition.

A key challenge in the original FRA proposed by Borwein and Wolkowicz is that finding a nonzero element in $SEP(L \cap \mathbb{S}^n_+)$ can be as difficult as solving the SDP itself. More practical approaches use a subset of $SEP(L \cap \mathbb{S}^n_+)$, in which finding exposing vectors is easier. In these cases, the resulting reformulation may not satisfy Slater's condition, and we obtain only *partial facial reduction*. Nevertheless, even partial facial reduction often reduces problem size and improves numerical stability in practice. Over the years, various FRAs have been developed based on different subsets of $SEP(L \cap \mathbb{S}^n_+)$. In Section 3, we propose a new FRA that leverages the structure of the underlying CO problems to find exposing vectors more effectively.

2.4 Polyhedral theory and linear algebra

Let $P = \{x \in \mathbb{R}^n \mid Ax \leq b\}$ be a polyhedron. An inequality of the form $c^T x \leq \delta$ is said to be *valid* for P if it holds for every $x \in P$. It is well known that the set of valid inequalities for P can be characterized by nonnegative combinations of the inequalities in $Ax \leq b$.

Lemma 1 ([9, Theorem 3.22]) *An inequality $c^T x \leq \delta$ is* valid *for P if and only if there exists $u \geq \mathbf{0}$ such that $u^T A = c^T$ and $u^T b \leq \delta$.*

The i-th inequality $a_i^T x \leq b_i$ in the system $Ax \leq b$ is called an *implicit equality* if $a_i^T x = b_i$ holds for every $x \in P$. The *affine hull* of P is the smallest affine set containing P. Denoting by $A^= x = b^=$ the system of all implicit equalities in $Ax \leq b$, the affine hull of P is given by

$$\text{aff } P = \{x \in \mathbb{R}^n \mid A^= x = b^=\}. \tag{11}$$

2.5 Explicit expression for $SEP(L \cap \mathcal{K})$

In [6,7] and the broader facial reduction literature, it is shown that when L is an affine set, the set $SEP(L \cap \mathcal{K})$ equals $L^\perp \cap \mathcal{K}^*$, where $\mathcal{K} \subseteq \mathbb{S}^n$ is a closed convex cone. To conclude this section, we derive an explicit expression for $SEP(L \cap \mathcal{K})$ when L is a general polyhedron. We also derive a more specialized expression for the case where L corresponds to the polyhedron in Shor's SDP relaxation, as defined in (3).

Lemma 2 *Let $\mathcal{K} \subseteq \mathbb{S}^n$ be a closed convex cone and $L = \{X \in \mathbb{S}^n \mid \mathcal{L}(X) \leq b\}$ for some linear operator $\mathcal{L} : \mathbb{S}^n \to \mathbb{R}^m$ and $b \in \mathbb{R}^m$. Suppose $L \cap \mathcal{K} \neq \emptyset$. Then,*

$$SEP(L \cap \mathcal{K}) = \left\{\mathcal{L}^*(y) \in \mathcal{K}^* \mid y \geq \mathbf{0},\, b^T y = 0\right\}. \tag{12}$$

Proof Let $\bar{X} \in L \cap \mathcal{K}$, and recall that $W \in SEP(L \cap \mathcal{K})$ if and only if it satisfies (8). For any $W \in SEP(L \cap \mathcal{K})$, we will show that $W \in \mathcal{K}^*$ and $W = \mathcal{L}^*(y)$ for some $y \geq \mathbf{0}$ with $b^T y = 0$.

First, suppose for contradiction that $W \notin \mathcal{K}^*$. Then there exists $\bar{Y} \in \text{relint}(\mathcal{K})$ such that $\langle W, \bar{Y} \rangle < 0$. However, (8) implies that

$$t\langle W, \bar{Y} \rangle = \langle W, t\bar{Y} \rangle \geq \langle W, \bar{X} \rangle \quad \text{for all } t > 0,$$

which becomes unbounded below as $t \to \infty$, contradicting the finiteness of $\langle W, \bar{X} \rangle$. Hence, $W \in \mathcal{K}^*$.

Next, from (8), for any $X \in L$ and $Y \in \text{relint}(\mathcal{K})$,

$$\langle W, X \rangle \leq \langle W, tY \rangle = t\langle W, Y \rangle \text{ for all } t > 0.$$

Taking $t \to 0$, we obtain $\langle W, X \rangle \leq 0$ for all $X \in L$. By LP strong duality, we have

$$0 \geq \max \{\langle W, X \rangle \mid X \in L\} = \min \left\{ b^T y \mid \mathcal{L}^*(y) = W,\, y \geq \mathbf{0} \right\}.$$

Therefore, there exists $y \geq \mathbf{0}$ such that $b^T y \leq 0$ and $W = \mathcal{L}^*(y)$. Moreover, since $\mathcal{L}(\bar{X}) \leq b$, $y \geq \mathbf{0}$, $\bar{X} \in \mathcal{K}$, and $W \in \mathcal{K}^*$,

$$b^T y \geq \langle \mathcal{L}(\bar{X}),\, y \rangle = \langle \bar{X}, \mathcal{L}^*(y) \rangle = \langle \bar{X}, W \rangle \geq 0,$$

which implies $b^T y = 0$. Hence, $W \in \mathcal{K}^*$ is in the set on the right-hand side of (12).

Conversely, suppose $W = \mathcal{L}^*(y) \in \mathcal{K}^*$ for some $y \geq \mathbf{0}$ with $b^T y = 0$. For any $X \in L$ and $Y \in \text{relint}(\mathcal{K})$,

$$\langle W, X \rangle = \langle \mathcal{L}^*(y), X \rangle = \langle \mathcal{L}(X),\, y \rangle \leq b^T y = 0 \leq \langle W, Y \rangle.$$

Therefore, $W \in SEP(L \cap \mathcal{K})$.

When L corresponds to the polyhedron in Shor's SDP relaxation, the expression for $SEP(L \cap \mathcal{K})$ can be further specified.

Lemma 3 *Let L be the polyhedron defined in (3), and let $\mathcal{K} \subseteq \mathbb{S}^{n+1}$ be a closed convex cone. Then,*

$$SEP(L \cap \mathcal{K}) = \left\{ \begin{bmatrix} -b^T y & \frac{1}{2}(y^T A - z^T) \\ \frac{1}{2}(A^T y - z) & Diag(z) \end{bmatrix} \in \mathcal{K}^* \,\middle|\, y \in \mathbb{R}^m_+,\ z \in \mathbb{R}^n \right\}. \quad (13)$$

Proof Applying Lemma 2 to the polyhedron L, where

$$\mathcal{L} = \begin{bmatrix} \mathcal{A} \\ Arrow \\ -Arrow \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} \mathbf{0} \\ e_0 \\ -e_0 \end{bmatrix},$$

we obtain that

$$
\begin{aligned}
&SEP(L \cap \mathcal{K}) \\
&= \left\{ \mathcal{A}^*(y) + Arrow^*(z^+) - Arrow^*(z^-) \in \mathcal{K}^* \;\middle|\; \begin{bmatrix} \mathbf{0} \\ e_0 \\ -e_0 \end{bmatrix}^T \begin{bmatrix} y \\ z^+ \\ z^- \end{bmatrix} = 0, \begin{bmatrix} y \\ z^+ \\ z^- \end{bmatrix} \geq \mathbf{0} \right\} \\
&= \left\{ \mathcal{A}^*(y) + Arrow^*(\hat{z}) \in \mathcal{K}^* \mid e_0^T \hat{z} = 0,\, \hat{z} \in \mathbb{R}^{n+1},\, y \in \mathbb{R}^m_+ \right\},
\end{aligned}
$$

where we define $\hat{z} = z^+ - z^-$. Writing $\hat{z} = (z_0, z) \in \mathbb{R} \times \mathbb{R}^n$, the constraint $e_0^T \hat{z} = 0$ implies $z_0 = 0$. Moreover, it is clear that

$$
Arrow^*(\hat{z}) = \begin{bmatrix} z_0 & -\frac{1}{2} z^T \\ -\frac{1}{2} z & \mathrm{Diag}(z) \end{bmatrix} \quad \text{and} \quad \mathcal{A}^*(y) = \begin{bmatrix} -b^T y & \frac{1}{2} y^T A \\ \frac{1}{2} A^T y & \mathbf{O} \end{bmatrix}.
$$

Substituting $z_0 = 0$ and combining the two terms yields the desired expression (13).

3 Affine FR

In this section, we introduce *affine FR* as a pre-processing technique to regularize SDP relaxations for CO problems. Affine FR is designed to be simple and efficient, in line with the principle of *simple and quickly* advocated by Andersen and Andersen [1]. As the effectiveness of pre-processing algorithms depends heavily on the structure of specific problem instances, a good pre-processing algorithm must balance computational cost with the degree of simplification achieved. If a pre-processing algorithm takes significant time without yielding meaningful reductions, it may have a detrimental impact on the overall computation time.

Affine FR exploits the fact that any proper affine subset in $\mathbb{R}^n$ containing the feasible region $F \subseteq \{0,1\}^n$ leads to facial reduction in the lifted matrix space. To see this, suppose that $F \subseteq H$, where H is an affine subspace of dimension $(r < n)$ defined by

$$
H := \left\{ x \in \mathbb{R}^n \mid U^T \left[\begin{smallmatrix} 1 \\ x \end{smallmatrix}\right] = \mathbf{0} \right\}, \text{ where } U \in \mathbb{R}^{(n+1)\times(n-r)}. \tag{14}
$$

Let $W := UU^T \in \mathbb{S}^{n+1}_+$. Since U has full column rank, we have rank $W = n - r$. We show that this matrix W satisfies $\langle W, Y \rangle = 0$ for every matrix $Y \in F_1$. Specifically, for any $x \in F$, let $\tilde{x} = \left[\begin{smallmatrix} 1 \\ x \end{smallmatrix}\right]$. It holds that

$$
\langle W, Y \rangle = \tilde{x}^T U U^T \tilde{x} = \left\| U^T \tilde{x} \right\|^2 = 0 \text{ for every } Y = \tilde{x}\tilde{x}^T \in F_1. \tag{15}
$$

Let $L \cap \mathbb{S}^{n+1}_+$ be any SDP relaxation for F. While W may not be an exposing vector for $L \cap \mathbb{S}^{n+1}_+$ in the strict sense, it still satisfies $\langle W, Y \rangle = 0$ for every $Y \in F_1$. Therefore, any SDP relaxation for F_1 can be strengthened by explicitly imposing the constraint $\langle W, Y \rangle = 0$, if it is not already implied.

Define $\bar{L} := L \cap \{Y \mid \langle W, Y\rangle = 0\}$. As $F_1 \subseteq \bar{L} \cap \mathbb{S}_+^{n+1}$, the set $\bar{L} \cap \mathbb{S}_+^{n+1}$ remains a valid SDP relaxation for F and is at least as strong as the original relaxation $L \cap \mathbb{S}_+^{n+1}$. Moreover, by construction, W is an exposing vector for $\bar{L} \cap \mathbb{S}_+^{n+1}$. Thus, we can perform an FR step to reformulate $\bar{L} \cap \mathbb{S}_+^{n+1}$, reducing the order of the matrix variable from $(n+1)$ to $(r+1)$.

To determine an affine set H in (14), we use the linear system of inequalities that define F. Since we assume that $F = P \cap \{0,1\}^n$, where $P = \{x \in \mathbb{R}^n \mid Ax \leq b\}$, it follows that $F \subseteq P$, and thus $F \subseteq \operatorname{aff} F \subseteq \operatorname{aff} P$. A practical advantage of using $\operatorname{aff} P$ is that affine FR can be applied without needing the explicit form of the SDP relaxation — it depends only on the linear system $Ax \leq b$. We describe an efficient subroutine to compute $\operatorname{aff} P$ in Section 3.1.

Now we provide a complete description of affine FR. Given $F = P \cap \{0,1\}^n$, where $P = \{x \in \mathbb{R}^n \mid Ax \leq b\}$, and an SDP relaxation for F, the affine FR procedure is as follows.

1. **Compute the affine hull:** Find a matrix U such that $H = \operatorname{aff} P$ is represented in the form given by (14).
2. **Construct a facial range vector:** Compute a matrix V whose columns form a basis for the null space of U^T. This matrix is the facial range vector associated with the exposing vector $W = UU^T$.
3. **Reformulate the SDP relaxation:** Replace $Y \in \mathbb{S}_+^{n+1}$ with a reduced form VRV^T, as demonstrated in (10).

Some remarks are in order.

- The SDP relaxation resulting from affine FR is not necessarily equivalent to the original relaxation. Specifically, it may be strictly stronger if the constraint $\langle W, Y\rangle = 0$ was not already implied in the original formulation. Nevertheless, we did not observe such strengthening in the numerical experiments presented in Section 5.2. For simplicity, we still refer to the new relaxation as a *reformulation* of the original. We say that affine FR *restores Slater's condition* if the reformulated relaxation satisfies Slater's condition.
- Unlike general FRAs, which typically apply facial reduction iteratively, affine FR performs only a single FR step.
- When $r = \dim(F)$, the affine space H coincides with the affine hull of F, and the face exposed by W is the smallest face of $\mathbb{S}_+^{n+1}$ containing $\bar{L} \cap \mathbb{S}_+^{n+1}$. In this case, affine FR performs complete facial reduction, restoring Slater's condition. See Lemma 4 and [41].
- Classical analytical FRAs, which guarantee complete facial reduction when applied to SDP relaxations of CO problems, can be viewed as a variant of affine FR, where $H = \operatorname{aff} P$ is replaced by $H = \operatorname{aff} F$ in the affine FR framework; see Section 4.1. However, computing $\operatorname{aff} F$ explicitly is often computationally challenging; in fact, it is NP-hard in general.[3] The core

[3] For example, given a set of positive integers $a_1, \ldots, a_n$, the subset sum problem asks whether there exists a subset $T \subseteq \{1, \ldots, n\}$ such that $\sum_{i \in T} a_i = \sum_{i \notin T} a_i$. The subset sum problem is known to be NP-hard; see [21, page 97]. Define the binary set $F = \{x \in \{0,1\}^n \mid \sum_{i=1}^n a_i x_i = \sum_{i=1}^n a_i(1 - x_i)\}$. The answer to the subset sum problem is NO if

idea of affine FR is to leverage an affine set that contains F to enable a computationally efficient approach. The trade-off is that this approach may yield only partial facial reduction, rather than a complete one.

– Even if Slater's condition is not restored, affine FR may still reduce the matrix size and decrease the singularity degree, thereby improving numerical stability.

We provide a concrete example to clarify affine FR next.

Example 1 *Consider the binary feasible set*

$$F := P \cap \{0,1\}^3 = \{(1,0,0),(0,1,0)\},$$

where $P = \{x \in [0,1]^3 \mid 2x_1 + x_2 \leq 2, x_1 + 2x_2 \leq 2, x_1 + x_2 \geq 1, x_3 \leq 0\}$. *The affine hull of* P *is*

$$\text{aff } P := \{x \in \mathbb{R}^3 \mid x_3 = 0\}.$$

The set F *and aff* P *are depicted in Figures 1 and 2. Applying affine FR, we obtain that*

$$U = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \; W = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}^T, \; \text{and } V = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus, we can reduce the order of the matrix variable in an SDP relaxation for F *from* 4 *to* 3.

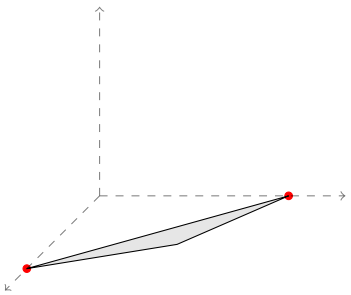

Fig. 1 The red dots constitute F, and the grey triangular area represents P.

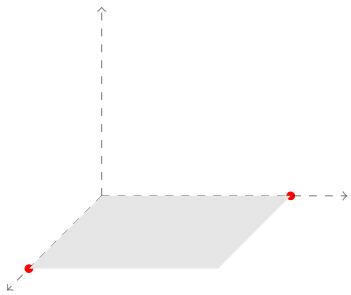

Fig. 2 The affine hull of P is represented by the (unbounded) gray area.

Note that the affine FR, which sets $H = \text{aff } P$, does not restore Slater's condition in the previous example. In contrast, as discussed in the earlier remarks, Slater's condition is restored in the continuation of the example below, where H is chosen to be aff F.

and only if $F = \emptyset$. Determining whether $F = \emptyset$ is equivalent to checking whether the affine hull of F is empty.

Example 2 *The affine hull of* $F := \{(1,0,0),(0,1,0)\} \subseteq \mathbb{R}^3$ *is given by*

$$\text{aff } F = \{x \in \mathbb{R}^3 \mid x_1 + x_2 = 1, x_3 = 0\}.$$

This is a one-dimensional affine subspace, as illustrated in Figures 3 and 4. From this affine hull, we obtain that

$$U = \begin{bmatrix} -1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \; W = UU^T = \begin{bmatrix} 1 & -1 & -1 & 0 \\ -1 & 1 & 1 & 0 \\ -1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \; \text{and } V = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Thus, we can reduce the order of the matrix variable in an SDP relaxation from 4 *to* 2*, and Slater's condition is restored.*

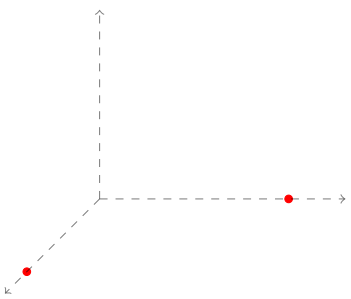

Fig. 3 The red dots constitute F.

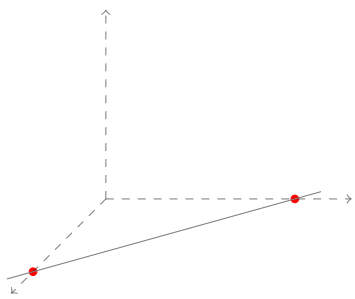

Fig. 4 The affine hull of F is represented by the line connecting the two red points.

3.1 Computing the affine hull

In this subsection, we discuss implementation details concerning the computation of the affine hull of a polyhedron P. Specifically, to determine aff P, it is necessary to identify all implicit equalities in the system $Ax \leq b$ defining P, as described in (11). In [12], Fukuda proposed a method that solves up to m LPs associated with P to detect all implicit equalities. However, this approach can be computationally expensive, particularly when the number of inequalities m is large. Here, we present a more efficient alternative based on a single LP formulation.

Assuming $P \neq \emptyset$, LP strong duality implies that the following primal and dual LPs share the same optimal value:

$$\max\{0 \mid Ax \leq b\} = \min\{b^T y \mid A^T y = \mathbf{0}, y \geq \mathbf{0}\}.$$

Note that any feasible solution x to the primal is optimal. Let y^* be any optimal solution to the dual. By the complementary slackness condition, it holds that $(b - Ax)^T y^* = 0$. Let $I := \{i \mid y_i^* > 0\}$ be the index set of positive entries in y^*. The primal constraints in $Ax \leq b$ associated with I are always

active, that is, for any $x \in P$, we have $a_i^T x = b_i$ for each $i \in I$. By definition, these constraints are implicit equalities of the system.

If the dual optimal solution y^* has the maximum number of nonzeros, then the corresponding set I captures all implicit equalities by the Goldman-Tucker Theorem; see [16, Corollary 2A]. In this case, the affine hull of P is given by

$$\text{aff } P = \{x \in \mathbb{R}^n \mid a_i^T x = b_i \text{ for every } i \in I\}.$$

There are several ways to compute such an optimal dual solution y^* with the maximum number of nonzeros. On approach is to use interior point methods, as shown by Güler and Ye [17]. Alternatively, y^* can be obtained by solving a single LP. Below we sketch a proof of this fact, based on [5, Exercise 3.27] and [29].

Consider the following LP:

$$\begin{aligned} \max\ & \mathbf{1}^T u \\ \text{subject to } & (u+v)^T \begin{bmatrix} A & b \end{bmatrix} = \mathbf{0}^T \\ & u, v \geq \mathbf{0} \\ & u \leq \mathbf{1}. \end{aligned} \tag{16}$$

Let y^* be an optimal dual solution with the maximum number of nonzero components. Since the dual optimal set is a cone, we can scale y^* so that all of its positive components are at least one, without altering its support. Define $u^* := \min\{y^*, \mathbf{1}\}$ and $v^* := y^* - u^*$. Then (u^*, v^*) is a feasible solution to the LP in (16), and the objective value $\mathbf{1}^T u^*$ equals the number of nonzero entries in y^*. Conversely, suppose that (u^*, v^*) is an optimal solution to (16). By optimality, each entry of u^* must be either 0 or 1. Otherwise, any fractional value in u^* could be increased (while adjusting v^* accordingly) to yield a better objective without violating feasibility. Additionally, if $u_i^* = 0$, it must follow that $v_i^* = 0$; otherwise, increasing u_i^* and decreasing v_i^* slightly would yield a strictly better solution. Therefore, the vector $y^* = u^* + v^*$ is a dual optimal solution with exactly $\mathbf{1}^T u^*$ nonzero components.

Thus, we can compute the affine hull of a polyhedron with n variables and m inequalities by solving the LP in (16), which has $2m$ variables and $n+1$ equality constraints. As the variable u is bounded above, the bounded-variable simplex method can be applied to solve (16). While the number of variables is doubled, the computational cost of solving (16) is negligible compared to that of solving an SDP relaxation.

Remark 1 The costs of solving (16) can be reduced as follows. Given a feasible solution $x \in P$, we can check whether each inequality $a_i^T x \leq b_i$ holds strictly. If so, it cannot be an implicit equality, and there is no need to include the variables u_i and v_i in (16). In many applications, it is relatively easy to generate feasible solutions, and they can be used to reduce the size of (16) substantially.

3.2 Application to the Lovász-Schrijver relaxation

Affine FR is not limited to Shor's relaxation — it can be applied to a broad class of SDP relaxations. In this section, we demonstrate the applicability of affine FR using the Lovász-Schrijver (LS) SDP relaxation [8,28] as a representative example.

Recall from (1) that $F = P \cap \{0,1\}^n$ is a binary set with $P = \{x \in \mathbb{R}^n \mid Ax \leq b\}$. Define the conic lift of P as

$$Q := \left\{ t \begin{bmatrix} 1 \\ x \end{bmatrix} \in \mathbb{R}^{n+1} \mid x \in P, t \geq 0 \right\}.$$

The LS SDP relaxation for F is given by

$$M_+(F) := \{Y \in \mathbb{S}_+^{n+1} \mid Arrow(Y) = e_0, Ye_i \in Q, Y(e_0 - e_i) \in Q \text{ for } i = 1, \ldots, n\}.$$

The LS SDP relaxation is typically larger and more computationally demanding than Shor's SDP relaxation. To address the computational difficulties, Burer and Vandenbussche [8] developed a specialized augmented Lagrangian method tailored to solve the LS relaxation efficiently. The FRA developed in this work constructs a more compact and numerically stable facially reduced formulation of $M_+(F)$. As a result, many algorithms applied to the facially reduced LS SDP relaxation are expected to perform better in terms of both computational efficiency and numerical stability.

To apply affine FR to $M_+(F)$, we begin by computing the affine hull of P, which yields an exposing vector $W \in \mathbb{S}_+^{n+1}$. According to (15), the lifted set F_1 (defined in (2)) is contained in the face of $\mathbb{S}_+^{n+1}$ exposed by W. This means that

$$M_+(F) \cap \{Y \in \mathbb{S}^{n+1} \mid \langle W, Y \rangle = 0\} \tag{17}$$

is also an SDP relaxation for F, and it is no weaker than $M_+(F)$. Moreover, any matrix Y in (17) can be written in the form of

$$Y = VRV^T \text{ for some } R \in \mathbb{S}_+^r,$$

where r is the dimension of the null space of W, and $V \in \mathbb{R}^{(n+1)\times r}$ is the corresponding facial range vector (see (10)). This allows us to express the facially reduced reformulation of (17) as

$$\tilde{M}_+(F) := \left\{ R \in \mathbb{S}_+^r \,\middle|\, \begin{array}{l} Arrow(VRV^T) = e_0,\ VRV^T e_i \in Q, \\ VRV^T(e_0 - e_i) \in Q \ \text{ for } i = 1, \ldots, n \end{array} \right\}.$$

The relationship between the original and reduced formulations is straightforward: $R \in \tilde{M}_+(F)$ if and only if VRV^T is feasible for (17).

4 A Theoretical Comparison

In this section, we present a theoretical comparison between affine FR and several existing approaches in the literature. Facial reduction involves identifying an exposing vector, as defined in (7). A central challenge in FRAs lies in efficiently computing such exposing vectors in practice.

Existing FRAs for SDP can generally be categorized into two main types:

1. Analytical Methods: There are tailored to specific problem classes and rely on analytical formulas to construct exposing vectors. They are highly efficient, as they involve little to no additional computation. However, their applicability is limited to problems with particular structural properties, and they often require substantial domain expertise to implement, which can make them less accessible to general users.
2. General-Purpose Methods: These are designed for a broad range of SDP problems. Notable examples include partial FR [33] and Sieve-SDP [49]. These methods are fully automatic, and their performance depends on the formulation of the SDP instance. However, they typically do not exploit the structural characteristics of the underlying CO problems when applied to its SDP relaxation. As a result, they may yield limited reduction, as observed in both our theoretical analysis and numerical experiments.

The key features and characteristics of these methods are summarized in Table 1.

Method	Applicability	Computational Cost	Effectiveness	User-Friendly
Analytical	Very Low	Zero	High	No
Sieve-SDP	High	Very Low	Low	Yes
Partial FR	High	Low	Low	Yes
Affine FR	High	Low	Medium	Yes

Table 1 Comparison of FRAs for CO problems

Each method in Table 1 has distinct advantages depending on the problem context. Analytical methods remain excellent choices for highly structured problems, while general-purpose methods are preferable when such structure is absent. This comparison aims to clarify these distinctions and help users select the most appropriate FRA for their specific needs.

For instance, when a problem exhibits substantial structure, it is worthwhile to investigate whether an analytical formula can be derived for its affine hull. Conversely, for general SDP problems lacking such structure, Sieve-SDP or Partial FR may be more appropriate. Finally, if the SDP problem arises as a relaxation of a combinatorial optimization problem, the affine FR approach developed in this work presents a compelling alternative.

The main result in this section is as follows.

Theorem 1 *Let $F := P \cap \{0,1\}^n$, where $P = \{x \in \mathbb{R}^n \mid Ax \leq b\}$, as defined in (1). We consider four FRAs applied to Shor's relaxation for F, as summarized*

in Table 2. Then,

$$r_A \leq r_P^+ \leq r_P \leq r_S = n + 1.$$

Matrix Size After Reduction	Method
r_A	Affine FR (Section 3)
r_p	Partial FR with cone $\mathcal{D}^*$ (Section 4.2.1)
r_p^+	Partial FR with cone $\mathcal{DD}^*$ (Section 4.2.2)
r_S	Sieve-SDP (Section 4.3)

Table 2 Matrix size notation after applying FRAs to Shor's relaxation

The proof of Theorem 1 is presented over the remainder of this section, through Corollary 2 and Lemma 10.

4.1 Analytical methods

Analytical methods operate similarly to affine FR, but differ in their choice of affine space H for deriving the exposing vector. Specifically, while affine FR computes the affine hull of the polyhedron P numerically, analytical methods aim to derive a closed-form expression for the affine hull of the binary feasible set F. In [47], Zhao et al. applied this analytical approach to construct SDP relaxations for the quadratic assignment problem that satisfy Slater's condition. Their method relies on the fact that the set of permutation matrices has a compact convex hull representation:

$$conv\, F = \{X \in [0,1]^{n\times n} \mid X\mathbf{1} = \mathbf{1}, X^T\mathbf{1} = \mathbf{1}, X \geq \mathbf{O}\}. \tag{18}$$

As $X \geq \mathbf{O}$ are not implicit equalities, the affine hull of F is fully determined by the equality constraints in (18). This structure enables an analytical derivation of aff F, facilitating complete facial reduction.

When such a closed-form expression for aff F can be derived, the analytical approach incurs virtually no computational cost, making it highly desirable. However, its applicability is quite limited. Most studies employing the analytical approach focus on variants of the quadratic assignment problem, including the graph partitioning problem [25,46], the min-cut problem [23], and the vertex separator problem [37]. Similar techniques have also been used in the quadratic cycle covering problem [10] and the quadratic shortest path problem [19] to restore Slater's condition for their SDP relaxations. In general, however, the binary feasible set F does not admit a compact affine hull description, and it is often NP-hard to derive an analytical formula. This limitation is a key motivation for developing the affine FR approach introduced in this work.

When applied to the CO problems mentioned above, affine FR always restores Slater's condition. This outcome follows directly from the fact that aff F = aff P holds for all these problems, together with Lemma 4 below.

Therefore, in situations where it is unclear whether a CO problem admits a tractable analytical formula for its affine hull, affine FR offers a robust and broadly applicable alternative.

Lemma 4 *Let $F = P \cap \{0,1\}^n$ for some polyhedron P. If aff F = aff P, then affine FR restores Slater's condition for any SDP relaxation for F.*

Proof Let $L \cap \mathbb{S}^{n+1}_+$ be an SDP relaxation for F, as defined in (3). Since aff F = aff P, we have $r := \dim F = \dim P$. Therefore, there exist $r+1$ affinely independent feasible points $v_1, \ldots, v_{r+1} \in F$. Define

$$\hat{Y} = \frac{1}{r+1} \sum_{i=1}^{r+1} \begin{bmatrix} 1 \\ v_i \end{bmatrix} \begin{bmatrix} 1 \\ v_i \end{bmatrix}^T .$$

By construction, $\hat{Y}$ is in $L \cap \mathbb{S}^{n+1}_+$ and has rank $r+1$. In addition, affine FR yields an exposing vector $W = UU^T$ of rank $n-r$, where $U \in \mathbb{R}^{(n+1)\times(n-r)}$ defines aff P, as in (14). Using the exposing vector, affine FR reformulates $L \cap \mathbb{S}^{n+1}_+$ as $\tilde{L} \cap \mathbb{S}^{r+1}_+$, where $\tilde{L} := \left\{ R \in \mathbb{S}^{r+1} \mid VRV^T \in L \right\}$, and $V \in \mathbb{R}^{(n+1)\times(r+1)}$ is the corresponding facial range vector, as described in (10). Since $\hat{Y} = V\hat{R}V^T$ for some $\hat{R} \in \tilde{L} \cap \mathbb{S}^{r+1}_+$ and $\hat{Y}$ has rank $r+1$, it follows that $\hat{R}$ is positive definite. Hence, the reduced formulation $\tilde{L} \cap \mathbb{S}^{r+1}_+$ satisfies Slater's condition, and affine FR restores Slater's condition.

4.2 Partial FR

Permenter and Parrilo [33] proposed a specialized FRA known as *partial facial reduction* (partial FR).[4] When applied to $L \cap \mathbb{S}^n_+$, partial FR aims to find, at each FR step, a *maximum-rank* matrix $W \in SEP(L \cap \mathcal{K})$, where $\mathcal{K}$ is a closed convex cone containing $\mathbb{S}^n_+$. Since $\mathcal{K} \supseteq \mathbb{S}^n_+$, it follows that $\mathcal{K}^* \subseteq \mathbb{S}^n_+$, and thus, by Lemma 2, $SEP(L \cap \mathcal{K}) \subseteq SEP(L \cap \mathbb{S}^n_+)$. Therefore, any nonzero matrix W identified in such an FR step serves as an exposing vector for $L \cap \mathbb{S}^n_+$. Following the general facial reduction framework described in Section 2.3, partial FR iteratively applies these steps until no further reduction is possible.

However, because $SEP(L \cap \mathcal{K})$ may not contain all exposing vectors for $L \cap \mathbb{S}^n_+$, partial FR may terminate early, failing to restore Slater's condition. On the other hand, partial FR benefits from low computational cost when $\mathcal{K}$ is chosen appropriately. Following the suggestions in [33], we consider two specific choices for $\mathcal{K}$:

1. The cone $\mathcal{D}^*$ is the set of symmetric matrices with nonnegative diagonal entries:
$$\mathcal{D}^* := \{X \in \mathbb{S}^n \mid X_{ii} \geq 0 \; \forall \, i\}.$$

[4] The term "partial facial reduction" also refers to the situation when facial reduction does not fully restore Slater's condition. The intended meaning is typically clear from the context.

Its dual cone $\mathcal{D}$ is the set of diagonal matrices with nonnegative diagonal entries:

$$\mathcal{D} = \{W \in \mathbb{S}^n \mid W_{ii} \geq 0,\, W_{ij} = 0 \,\forall\, i \neq j\}\,.$$

2. The cone $\mathcal{DD}^*$ is defined as

$$\mathcal{DD}^* := \{X \in \mathbb{S}^n \mid X_{ii} \geq 0,\, X_{ii} + X_{jj} \pm X_{ij} \geq 0 \,\forall\, i, j\}\,. \tag{19}$$

Its dual cone $\mathcal{DD}$ is the set of symmetric diagonally dominant matrices:

$$\mathcal{DD} = \left\{ W \in \mathbb{S}^n \;\middle|\; W_{ii} \geq \sum_{j \neq i} |W_{ij}| \,\forall\, i \right\}. \tag{20}$$

Since the dimensions of the cones above are always evident from the context, we omit their dimensional superscripts to simplify the notation.

We discuss these two choices in detail in the following subsections. Throughout this section, we adopt the same notation as in Section 2.2, unless stated otherwise. Specifically, let $F = P \cap \{0,1\}^n$, where $P = \{x \in \mathbb{R}^n \mid Ax \leq b\}$, and let $L \cap \mathbb{S}^{n+1}_+$ denote Shor's relaxation for F, where L is defined in (3). In addition, we assume that Assumption 1 holds throughout this section.

4.2.1 Non-negative diagonal matrices

We first simplify the expression for $SEP(L \cap \mathcal{K})$ when $\mathcal{K} = \mathcal{D}^*$.

Lemma 5 *It holds that*

$$SEP(L \cap \mathcal{D}^*) = \left\{ \begin{bmatrix} 0 & \boldsymbol{0}^T \\ \boldsymbol{0} & Diag\left(A^T y\right) \end{bmatrix} \middle| A^T y \geq \boldsymbol{0},\, b^T y = 0,\, y \in \mathbb{R}^m_+ \right\}. \tag{21}$$

Proof By Lemma 3, we have

$$SEP(L \cap \mathcal{D}^*) = \left\{ \begin{bmatrix} -b^T y & \frac{1}{2}(y^T A - z^T) \\ \frac{1}{2}(A^T y - z) & \mathrm{Diag}(z) \end{bmatrix} \in \mathcal{D} \middle| y \in \mathbb{R}^m_+, z \in \mathbb{R}^n \right\}.$$

Since $\mathcal{D}$ consists of diagonal matrices with nonnegative diagonal entries, the above expression simplifies to

$$SEP(L \cap \mathcal{D}^*) = \left\{ \begin{bmatrix} -b^T y & \mathbf{0}^T \\ \mathbf{0} & \mathrm{Diag}(A^T y) \end{bmatrix} \middle| b^T y \leq 0,\, A^T y \geq \mathbf{0},\, y \in \mathbb{R}^m_+ \right\}. \tag{22}$$

By Assumption 1, the polyhedron P is nonempty and $x \geq \mathbf{0}$ for every $x \in P$. Fix any $x^* \in P$. For any $y \in \mathbb{R}^m_+$ satisfying $b^T y \leq 0$ and $A^T y \geq \mathbf{0}$, we have that

$$0 \leq y^T A x^* \leq y^T b \leq 0. \tag{23}$$

As a result, (22) is equivalent to (21), which completes the proof.

In the next lemma, we present an explicit form of the exposing vectors generated by partial FR with $\mathcal{K} = \mathcal{D}^*$. We also show that the algorithm can terminate after at most one FR step. The key observation underlying the proof is that, after one FR step, the new matrix variable R is obtained from the original matrix variable Y by removing the rows and columns corresponding to the variables that are fixed at zero in P.

Lemma 6 *For each $i \in \{1, \ldots, n\}$, define*

$$W^i := \begin{bmatrix} 0 & \boldsymbol{0}^T \\ \boldsymbol{0} & e_i e_i^T \end{bmatrix} \in \mathbb{S}^{n+1},$$

where $e_i \in \mathbb{R}^n$ is the i-th standard basis vector. Let

$$I_+ := \{i \in \{1, \ldots, n\} \mid x_i = 0 \text{ for all } x \in P\},$$

and define $W^* = \sum_{i \in I_+} W^i$. Then the following statements hold:

1. *$W^* \in SEP(L \cap \mathcal{D}^*)$.*
2. *W^* has the maximum rank among all matrices in $SEP(L \cap \mathcal{D}^*)$.*
3. *When applied to $L \cap \mathbb{S}_+^{n+1}$, partial FR with $\mathcal{K} = \mathcal{D}^*$ can terminate after at most one FR step.*

Proof 1. If $i \in I_+$, then $x_i = 0$ for every $x \in P$. LP strong duality implies that

$$\min\{b^T y \mid A^T y = e_i, y \geq \mathbf{0}\} = \max\{x_i \mid Ax \leq b\} = 0. \tag{24}$$

Therefore, there exists $y \in \mathbb{R}_+^m$ such that $A^T y = e_i$ and $b^T y = 0$, which implies $W^i \in SEP(L \cap \mathcal{D}^*)$ by Lemma 5. As $SEP(L \cap \mathcal{D}^*)$ is a polyhedral cone, it follows $W^* = \sum_{i \in I_+} W^i \in SEP(L \cap \mathcal{D}^*)$.

2. If $i \notin I_+$, then $x_i^* > 0$ for some $x^* \in P$. In this case, any $y \in \mathbb{R}_+^m$ satisfying $A^T y \geq \mathbf{0}$ and $b^T y = 0$ must also satisfy (23), and therefore $(A^T y)_i = 0$. Consequently, every matrix $W \in SEP(L \cap \mathcal{D}^*)$ must have zero entries in the i-th row and column whenever $i \notin I_+$ (with indices starting from 0). Therefore, the only positions where nonzero entries can appear in W are those indexed by $i \in I_+$. Since W^* is constructed to be nonzero exactly in these positions, it achieves the maximum possible rank among all matrices in $SEP(L \cap \mathcal{D}^*)$.
3. If $SEP(L \cap \mathcal{D}^*) = \{\mathbf{O}\}$, then partial FR with $\mathcal{K} = \mathcal{D}^*$ terminates immediately. Otherwise, the algorithm generates a maximum-rank exposing vector in $SEP(L \cap \mathcal{D}^*)$. Without loss of generality, assume that W^* is selected to be the exposing vector[5]. Let $V \in \mathbb{R}^{(n+1)\times(n+1-|I_+|)}$ be the associated facial range vector, obtained by removing the i-th column of the identity matrix I_{n+1} for each $i \in I_+$ (with column indices starting from 0). Using the reformulation in (10), the matrix variable R in the reduced problem is obtained from Y by removing the rows and columns corresponding to

[5] Any other maximum-rank exposing vector in this cone has the same null space as W^*, leading to the same facial reduction.

indices $i \in I_+$. Let $\tilde{P}$ be the polyhedron obtained by eliminating from P all variables x_i with $i \in I_+$, and define $\tilde{F} := \tilde{P} \cap \{0,1\}^{n-|I_+|}$. Then the reformulation $\tilde{L} \cap \mathbb{S}_+^{n+1-|I_+|}$ coincides with Shor's relaxation for $\tilde{F}$. Since $\tilde{P}$ contains no variables fixed at zero, we have $SEP(\tilde{L} \cap \mathcal{D}^*) = \{\mathbf{O}\}$, and partial FR with $\mathcal{K} = \mathcal{D}^*$ terminates after this step.

We remark that for each $i \in I_+$, the variable x_i is fixed at zero with respect to the polyhedron P. However, it is possible that $x_i > 0$ for some $x \in P$, while $x_i = 0$ for all $x \in F$. Consequently, this approach does not detect all variables that are fixed at zero in the binary feasible set F. To compute the maximum-rank exposing vector W^* in Lemma 6, one may apply the technique described in Section 3.1. An alternative method is proposed in [33].

Corollary 1 *It holds that $r_A \leq r_P$.*

Proof By definition of the index set I_+, it holds that $P \subseteq \{x \in \mathbb{R}^n \mid x_i = 0 \text{ for } i \in I_+\}$. Therefore, $\dim(P) \leq \dim(\{x \in \mathbb{R}^n \mid x_i = 0 \text{ for } i \in I_+\}) = n - |I_+|$. By construction, $r_A = \dim(P) + 1$ and $r_P = n + 1 - |I_+|$. Therefore, $r_A \leq r_P$, as claimed.

We illustrate the ideas presented in the preceding results with the following examples.

Example 3 *Let P be the polyhedron defined by the linear system $Ax \leq b$, where*

$$A = \begin{bmatrix} 1 & 1 \\ -1 & -1 \\ 1 & 0 \\ -1 & 0 \\ 0 & -1 \end{bmatrix} \text{ and } b = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

It is clear that $P = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 + x_2 = 1,\, x_1 = 0,\, x_2 \geq 0\} = \{(0,1)\}$. Let $F := \{0,1\}^2 \cap P = \{(0,1)\}$. The corresponding Shor's SDP relaxation is $L \cap \mathbb{S}_+^3$, where

$$\begin{aligned} L &= \left\{ Y = \begin{bmatrix} Y_{00} & Y_{01} & Y_{02} \\ Y_{10} & Y_{11} & Y_{12} \\ Y_{20} & Y_{21} & Y_{22} \end{bmatrix} \in \mathbb{S}^3 \;\middle|\; \begin{array}{c} Y_{00} = Y_{01} + Y_{02},\, Y_{01} = 0,\, Y_{02} \geq 0, \\ Y_{00} = 1,\, Y_{11} = Y_{01},\, Y_{22} = Y_{02} \end{array} \right\} \\ &= \left\{ \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & Y_{12} \\ 1 & Y_{12} & 1 \end{bmatrix} \;\middle|\; Y_{12} \in \mathbb{R} \right\}. \end{aligned}$$

Now apply partial FR with $\mathcal{D}^*$ to $L \cap \mathbb{S}^3_+$. Lemma 5 implies that

$$\begin{aligned} SEP(L \cap \mathcal{D}^*) &= \left\{ \begin{bmatrix} 0 & 0 & 0 \\ 0 & y_1 - y_2 + y_3 - y_4 & 0 \\ 0 & 0 & y_1 - y_2 - y_5 \end{bmatrix} \middle| \begin{array}{l} y_1 - y_2 + y_3 - y_4 \geq 0, \\ y_1 - y_2 - y_5 \geq 0, \\ y_1 - y_2 = 0,\ y \geq \boldsymbol{0} \end{array} \right\} \\ &= \left\{ \begin{bmatrix} 0 & 0 & 0 \\ 0 & y_3 - y_4 & 0 \\ 0 & 0 & -y_5 \end{bmatrix} \middle| y_3 - y_4 \geq 0,\ -y_5 \geq 0,\ y \geq \boldsymbol{0} \right\} \\ &= \left\{ \begin{bmatrix} 0 & 0 & 0 \\ 0 & t & 0 \\ 0 & 0 & 0 \end{bmatrix} \middle| t \geq 0 \right\}. \end{aligned}$$

Therefore, the matrix

$$W = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

is a maximum-rank exposing vector, consistent with the description of W^* in Lemma 6. Let

$$V = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \in \mathbb{R}^{3\times 2}.$$

be a corresponding facial range vector. Then, the matrix variable $Y \in \mathbb{S}^3_+$ can be represented as VRV^T, where $R = \begin{bmatrix} Y_{00} & Y_{02} \\ Y_{20} & Y_{22} \end{bmatrix}$. Indeed, the fact that $x_1 = 0$ for every $x \in P$ is reflected in the above process: R is obtained from Y by removing the row and column corresponding to index $i = 1$.

The resulting facially reduced problem is $\tilde{L} \cap \mathbb{S}^2_+$, where

$$\tilde{L} = \left\{ R = \begin{bmatrix} Y_{00} & Y_{02} \\ Y_{20} & Y_{22} \end{bmatrix} \in \mathbb{S}^2 \middle| \begin{array}{l} Y_{00} = Y_{02},\ Y_{02} \geq 0, \\ Y_{00} = 1,\ Y_{22} = Y_{02} \end{array} \right\} = \left\{ \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \right\}.$$

Let $\tilde{P} = \{x_2 \in \mathbb{R} \mid x_2 = 1,\ x_2 \geq 0\}$ be the polyhedron obtained from P by eliminating x_1, which is fixed at 0 in P. It is straightforward to verify that $\tilde{L} \cap \mathbb{S}^2_+$ is Shor's SDP relaxation for $\tilde{F} = \tilde{P} \cap \{0, 1\}$. Applying partial FR with $\mathcal{D}^*$ to this reduced problem yields

$$SEP(\tilde{L} \cap \mathcal{D}^*) = \left\{ \begin{bmatrix} 0 & 0 \\ 0 & y_1 - y_2 - y_5 \end{bmatrix} \middle| \begin{array}{l} y_1 - y_2 - y_5 \geq 0, \\ y_1 - y_2 = 0,\ y \geq \boldsymbol{0} \end{array} \right\} = \{\mathbf{O}\}.$$

This shows that there is no reduction in the second iteration, and the algorithm terminates with the final matrix dimension $r_P = 2$.

Example 4 *We now apply affine FR to the same set F as in Example 3. The affine hull of P is given by*

$$\mathit{aff}\ P = \{x \in \mathbb{R}^2 \mid x_1 + x_2 = 1, x_1 = 0\}.$$

This yields a rank-2 exposing vector

$$W = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ -1 & 0 \end{bmatrix}^T = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 1 \end{bmatrix}.$$

An associated facial range vector is

$$V = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \in \mathbb{R}^{3\times 1},$$

which implies that the reduced matrix dimension is $r_A = 1$.

4.2.2 Diagonally dominant matrices

In analogy to Lemma 5, we first simplify the expression for $SEP(L \cap \mathcal{K})$ when $\mathcal{K} = \mathcal{DD}^*$. Recall that $|u| = (|u_1|, \ldots, |u_n|)$ represents the componentwise absolute value of u.

Lemma 7 *It holds that*

$$SEP(L \cap \mathcal{DD}^*) = \left\{ \begin{bmatrix} -b^T y & \frac{1}{2}(y^T A - z^T) \\ \frac{1}{2}(A^T y - z) & Diag(z) \end{bmatrix} \middle| \begin{array}{c} -b^T y \geq \frac{1}{2}\boldsymbol{1}^T |A^T y - z| \\ z \geq \frac{1}{2}|A^T y - z| \\ y \in \mathbb{R}^m_+,\ z \in \mathbb{R}^n \end{array} \right\}.$$

Proof By Lemma 3, we have

$$SEP(L \cap \mathcal{DD}^*) = \left\{ \begin{bmatrix} -b^T y & \frac{1}{2}(y^T A - z^T) \\ \frac{1}{2}(A^T y - z) & \mathrm{Diag}(z) \end{bmatrix} \in \mathcal{DD} \middle| y \in \mathbb{R}^m_+, z \in \mathbb{R}^n \right\}.$$

Recall that $\mathcal{DD}$ is the set of diagonally dominant matrices and it is defined in (20). The above condition that the matrix is diagonally dominant is equivalent to the inequalities

$$-b^T y \geq \frac{1}{2}\mathbf{1}^T |A^T y - z| \quad \text{and} \quad z \geq \frac{1}{2}|A^T y - z|, \tag{25}$$

which completes the proof.

Given a polyhedron $P = \{x \in \mathbb{R}^n \mid Ax \leq b\}$, we partition $\{1, \ldots, n\}$ into the following mutually exclusive index sets:

$$\begin{aligned} I_+ &:= \{i \in \{1, \ldots, n\} \mid x_i = 0 \text{ for all } x \in P\}, \\ I_- &:= \{i \in \{1, \ldots, n\} \mid x_i = 1 \text{ for all } x \in P\}, \\ I_0 &:= \{i \in \{i, \ldots, n\} \mid x_i \in (0, 1) \text{ for some } x \in P\}. \end{aligned}$$

The following lemma shows that the row and column of an exposing vector corresponding to an index $i \in I_0$ must be zero.

Lemma 8 *If* $W \in SEP(L \cap \mathcal{DD}^*)$*, then* $W_{ij} = W_{ji} = 0$ *for all* $i \in I_0$ *and* $j \in \{0, 1, \ldots, n\}$*. Moreover, if* $I_- = I_+ = \emptyset$*, then* $SEP(L \cap \mathcal{DD}^*) = \{\boldsymbol{O}\}$.

Proof By Lemma 7, if $W \in SEP(L \cap \mathcal{DD}^*)$, then there exist $y \in \mathbb{R}^m_+$ and $z \in \mathbb{R}^n$ satisfying (25). Let $w := A^T y$. Since $z \geq \frac{1}{2}|w - z|$, we have $4z_i^2 \geq (w_i - z_i)^2$ for all i, which is equivalent to

$$(w_i - 3z_i)(w_i + z_i) \leq 0.$$

If $w_i < 0$, then $w_i - 3z_i < 0$, and we must have $w_i + z_i \geq 0$ and

$$|w_i - z_i| = z_i - w_i \geq -2w_i.$$

For any $x \in P$,

$$\begin{aligned}
-b^T y &\geq \frac{1}{2}\mathbf{1}^T|w - z| \\
&= \sum_{i:w_i<0} \frac{1}{2}|w_i - z_i| + \sum_{i:w_i\geq 0} \frac{1}{2}|w_i - z_i| \\
&\geq \sum_{i:w_i<0} \frac{1}{2}|w_i - z_i| \\
&\geq \sum_{i:w_i<0} -w_i \\
&\geq \sum_{i:w_i<0} -w_i x_i + \sum_{i:w_i=0} -w_i x_i + \sum_{i:w_i>0} -w_i x_i \\
&= -y^T A x \\
&\geq -b^T y.
\end{aligned}$$

Here, the fourth inequality holds due to Assumption 1. Since the first and last expression in this chain are equal, all inequalities must hold at equality. In particular, equality in

$$\sum_{i:w_i<0} -w_i \geq \sum_{i:w_i<0} -w_i x_i + \sum_{i:w_i=0} -w_i x_i + \sum_{i:w_i>0} -w_i x_i$$

implies

$$x_i = \begin{cases} 1, & \text{if } w_i < 0 \\ 0, & \text{if } w_i > 0 \end{cases} \quad \text{for all } x \in P.$$

Hence, if $x_i \in (0, 1)$ for some $x \in P$, then $w_i = 0$. Moreover, equality in

$$\sum_{i:w_i<0} \frac{1}{2}|w_i - z_i| + \sum_{i:w_i\geq 0} \frac{1}{2}|w_i - z_i| \geq \sum_{i:w_i<0} \frac{1}{2}|w_i - z_i|$$

implies that $w_i = z_i$ whenever $w_i \geq 0$. Therefore, for $i \in I_0$, we have $z_i = w_i = 0$, which implies $W_{ij} = W_{ji} = 0$ for all $j \in \{0, 1, \ldots, n\}$. Finally, if $I_- = I_+ = \emptyset$, then $I_0 = \{1, \ldots, n\}$, so every entry of W is zero except possibly $W_{00} = -b^T y$. However, since $z_i = w_i = 0$ for all $i \in \{1, \ldots, n\}$, every term in the chain of inequalities equals zero, implying $-b^T y = 0$. Therefore, $W = \mathbf{O}$ is the only element in $SEP(L \cap \mathcal{DD}^*)$.

In analogy to Lemma 6, the following result provides an explicit form of the exposing vectors generated by partial FR with $\mathcal{K} = \mathcal{DD}^*$. We also show that the algorithm can terminate after at most one FR step. The key observation is that, after a single FR step, the new matrix variable R is obtained from the original matrix variable Y by removing the rows and columns corresponding to the variables that are fixed at zero or one in P.

Without loss of generality, we assume that

$$I_- = \{1, \ldots, p\} \quad \text{and} \quad I_+ = \{p+1, \ldots, p+q\}.$$

When $I_- = \emptyset$, we set $p = 0$; similarly, when $I_+ = \emptyset$, we set $q = 0$.

Lemma 9 *Define the block-diagonal matrix $W^* \in \mathbb{S}_+^{n+1}$ as follows:*

$$W^* = \left[\begin{array}{cc|c|c} p & -\boldsymbol{1}^T & & \\ -\boldsymbol{1} & I_p & & \\ \hline & & I_q & \\ \hline & & & \boldsymbol{O}_{n-p-q} \end{array}\right], \tag{26}$$

where I_p and I_q are identity matrices of order p and q, respectively, and $\mathbf{O}_{n-p-q}$ is the zero matrix of order $n-p-q$.

It holds that

1. *$W^* \in SEP(L \cap \mathcal{DD}^*)$, and rank $W^* = p + q$.*
2. *W^* has the maximum rank among all matrices in $SEP(L \cap \mathcal{DD}^*)$.*
3. *When applied to $L \cap \mathbb{S}_+^{n+1}$, partial FR with $\mathcal{K} = \mathcal{DD}^*$ can terminate after at most one FR step.*

Proof 1. For each $i \in I_-$, since $x_i = 1$ for all $x \in P$, LP strong duality implies

$$\max\{-b^T y \mid A^T y = -e_i,\ y \geq \mathbf{0}\} = \min\{x_i \mid Ax \leq b\} = 1.$$

Therefore, there exists $y^i \in \mathbb{R}_+^m$ such that $A^T y^i = -e_i$ and $b^T y = -1$. Similarly, for $i \in I_+$, since $x_i = 0$ for all $x \in P$, by (24), there exists $y^i \in \mathbb{R}_+^m$ such that $A^T y^i = e_i$ and $b^T y = 0$. Define $y := \sum_{i \in I_- \cup I_+} y^i$ and $w := A^T y$. Then,

$$w_i = \begin{cases} -1 & \text{if } i \in I_- \\ 1 & \text{if } i \in I_+ \\ 0 & \text{otherwise.} \end{cases}$$

and $b^T y = -p$. Let $z := |w| \in \mathbb{R}^n$ be the componentwise absolute value of w. It is straightforward to verify that $-b^T y = p = \frac{1}{2}\mathbf{1}^T |w - z|$ and $z \geq \frac{1}{2}|w - z|$. Therefore, by Lemma 7,

$$W^* = \begin{bmatrix} -b^T y & \frac{1}{2}(w-z)^T \\ \frac{1}{2}(w-z) & \mathrm{Diag}(z) \end{bmatrix} \in SEP(L \cap \mathcal{DD}^*).$$

To prove that rank $W^* = p + q$, it suffices to show

$$\operatorname{rank} \begin{bmatrix} p & -\mathbf{1}^T \\ -\mathbf{1}^T & I_p \end{bmatrix} = p.$$

The rank of this matrix is at least p as it contains the identity matrix I_p as a submatrix, while the rank is at most p since the all-ones vector in $\mathbb{R}^{p+1}$ is in the null space of this matrix.

2. Let $W \in SEP(L \cap \mathcal{DD}^*)$ be arbitrary. By Lemma 8, the i-th row and column of W must be zero for $i = p+q+1, \ldots, n$ (with indices starting from 0). Therefore, W must be of the form

$$W = \begin{bmatrix} \bar{W} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix} \in \mathbb{S}_+^{n+1}$$

for some $\bar{W} \in \mathbb{S}_+^{p+q+1}$. It follows that rank $W \leq p+q+1$. Suppose, for the sake of contradiction, that rank $W = p+q+1$, i.e., $\bar{W}$ is positive definite. Then a corresponding facial range vector $V \in \mathbb{R}^{(n+1)\times(n-p-q)}$ is given by

$$V = \begin{bmatrix} \mathbf{O} \\ I_{n-p-q} \end{bmatrix}.$$

Now consider the facially reduced set

$$\tilde{L} := \{R \in \mathbb{S}^{n-p-q} \mid VRV^T \in L\},$$

where R is the reduced matrix variable. Observe that the first row of V is zero, and hence for any $R \in \mathbb{S}^{n-p-q}$, we have $(VRV^T)_{00} = 0$. However, the original arrow constraint in Shor's relaxation requires $Y_{00} = (VRV^T)_{00} = 1$. This implies $\tilde{L} = \emptyset$, contradicting to the assumption that $P \neq \emptyset$ in Assumption 1. Therefore, no matrix in $SEP(L \cap \mathcal{DD}^*)$ can have rank greater than $p+q$, and thus W^* has the maximum rank among all matrices in $SEP(L \cap \mathcal{DD}^*)$.

3. If $SEP(L \cap \mathcal{DD}^*) = \{\mathbf{O}\}$, then partial FR with $\mathcal{K} = \mathcal{DD}^*$ terminates immediately. Otherwise, the algorithm generates a maximum-rank exposing vector in $SEP(L \cap \mathcal{DD}^*)$. Without loss of generality, assume that W^* is selected to be the exposing vector. Define $r := n-p-q$. A facial range vector $V \in \mathbb{R}^{(n+1)\times(r+1)}$ associated with W^* is given by

$$V = \begin{bmatrix} \mathbf{1}_{p+1} & \mathbf{O} \\ \mathbf{0}_q & \mathbf{O} \\ \mathbf{0}_r & I_r \end{bmatrix}.$$

Let $R \in \mathbb{S}^{r+1}$ be the reduced matrix variable in the facially reduced formulation. We now derive the constraints of the facially reduced formulation. Recall that the original polyhedron P is defined by the linear system $Ax \leq b$, where the i-th constraint is given by $a_i^T x \leq b_i$. Let $\tilde{A} \in \mathbb{R}^{m\times r}$ be the submatrix of A consisting of its last r columns, and denote by $\tilde{a}_i^T$ the i-th row of $\tilde{A}$. Define the modified right-hand side $\tilde{b} \in \mathbb{R}^m$ by $\tilde{b}_i = b_i - \sum_{j\in I_-}(a_i)_j$, for $i = 1, \ldots, m$. Then the data matrix corresponding to the i-th constraint in (4) becomes

$$V^T \begin{bmatrix} -b_i & \frac{1}{2}a_i^T \\ \frac{1}{2}a_i & \mathbf{O} \end{bmatrix} V = \begin{bmatrix} -\tilde{b}_i & \frac{1}{2}\tilde{a}_i^T \\ \frac{1}{2}\tilde{a}_i & \mathbf{O} \end{bmatrix} \in \mathbb{S}^{r+1}.$$

Under this reduction, the arrow constraint (5) transforms to $\tilde{Arrow}(R) = \tilde{e}_0$ with $\tilde{Arrow} : \mathbb{S}^{r+1} \to \mathbb{R}^{r+1}$ given by

$$\tilde{Arrow}(R) := \begin{bmatrix} R_{00} \\ R_{11} - \frac{1}{2}(R_{01} + R_{10}) \\ \vdots \\ R_{rr} - \frac{1}{2}(R_{0r} + R_{r0}) \end{bmatrix} \in \mathbb{R}^{r+1},$$

where $\tilde{e}_0$ is the first standard basis vector in $\mathbb{R}^{r+1}$. Let $\tilde{L}$ be the set defined by

$$\left\langle \begin{bmatrix} -\tilde{b}_i & \frac{1}{2}\tilde{a}_i^T \\ \frac{1}{2}\tilde{a}_i & \mathbf{O} \end{bmatrix}, R \right\rangle \leq 0 \quad \text{for } i = 1, \ldots, m$$

and $\tilde{Arrow}(R) = \tilde{e}_0$. Then the facially reduced reformulation of Shor's relaxation $L \cap \mathbb{S}_+^{n+1}$ is $\tilde{L} \cap \mathbb{S}_+^{r+1}$. On the other hand, $\tilde{L} \cap \mathbb{S}_+^{r+1}$ is exactly Shor's relaxation of the binary set

$$\tilde{F} := \{0,1\}^r \cap \tilde{P}, \text{ where } \tilde{P} := \{y \in \mathbb{R}^r \mid \tilde{A}y \leq \tilde{b}\}.$$

Note that

$$y \in \tilde{F} \text{ if and only if } \begin{bmatrix} \mathbf{1} \\ \mathbf{0} \\ y \end{bmatrix} \in F.$$

By construction, none of the variables in $\tilde{P}$ is fixed at zero or one. Therefore, Lemma 8 implies that $SEP(\tilde{L} \cap \mathbb{S}_+^{r+1}) = \{\mathbf{O}\}$, and partial FR with $\mathcal{K} = \mathcal{DD}^*$ terminates after this step.

Corollary 2 *It holds that $r_A \leq r_P^+ \leq r_P$.*

Proof The proof follows the same reasoning as in Corollary 1. Define two affine sets:

$$P' := \left\{ x \in \mathbb{R}^n \,\middle|\, \begin{array}{l} x_i = 0 \text{ for } i \in I_+ \\ x_i = 1 \text{ for } i \in I_- \end{array} \right\} \text{ and } P'' := \{x \in \mathbb{R}^n \mid x_i = 0 \text{ for } i \in I_+\}.$$

By definition of the index sets I_+ and I_-, it holds that $P \subseteq P' \subseteq P''$. Therefore, $\dim(P) \leq \dim(P') \leq \dim(P'')$. By construction, $r_A = \dim(P) + 1$, $r_P^+ = n + 1 - |I_+| - |I_-| = \dim(P') + 1$ and $r_P = n + 1 - |I_+| = \dim(P'') + 1$. Therefore, $r_A \leq r_P^+ \leq r_P$, as claimed.

Note that the proof of Lemma 9 also demonstrates that a maximum-rank exposing vectors for $L \cap \mathcal{DD}^*$ can be computed by solving a single LP. Therefore, its computational complexity is comparable to that of affine FR.

Corollary 2 shows that, for any linear system $Ax \leq b$ defining the binary set F, affine FR achieves at least as much reduction as partial FR with $\mathcal{D}^*$ or $\mathcal{DD}^*$ applied to Shor's SDP relaxation. This suggests that leveraging structural knowledge of the underlying binary set F can aid in identifying effective exposing vectors.

Using more accurate inner approximations of $\mathbb{S}^n_+$, e.g., factor-width-k matrices, could potentially enhance the reduction obtained via partial FR. However, identifying an exposing vector from the corresponding set becomes significantly more challenging. Even the cone of diagonally dominant matrices $\mathcal{DD}$ involves $\mathcal{O}(n^2)$ variables in general, making it impractical for problems with thousands of variables. Since affine FR requires only n variables, we do not compare it here with more accurate inner approximations used in partial FR.

Remark 2 All results in this section extend naturally to mixed-binary linear programming problems. For example, if F is a mixed-binary feasible set defined as the feasible region of (30), then partial FR with $\mathcal{D}^*$ applied to Shor's SDP relaxation detects only binary variables fixed at zero in P, while leaving continuous variables unaffected. Similarly, when partial FR with $\mathcal{DD}^*$ is applied to Shor's relaxation, it eliminates only binary variables fixed at zero or one. The proof of this extension follows the same reasoning as in Lemmas 6 and 9.

4.3 Sieve-SDP

In [49], Zhu et al. propose a specialized FRA called *Sieve-SDP*. Their method aims to efficiently identify certain structural patterns in the constraint matrices that allow for low-cost extraction of exposing vectors. The facial reduction component of Sieve-SDP can be summarized as follows: for each constraint of the form $\langle A_i, Y\rangle = b_i$ or $\langle A_i, Y\rangle \leq b_i$, the algorithm checks whether the matrix A_i can be permuted into the block-diagonal form:

$$A_i = \begin{bmatrix} D_i & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{bmatrix} \text{ with } D_i \in \mathbb{S}^k_{++}. \tag{27}$$

If $b_i = 0$, then A_i is an exposing vector, and facial reduction can be applied by eliminating the rows and columns of the matrix variable Y corresponding to the block D_i. The algorithm terminates when no such constraint can be found. Sieve-SDP effectively searches for specific elements within the set of exposing vectors and relies only on an incomplete Cholesky factorization to verify the positive definiteness of D_i. This makes Sieve-SDP an extremely fast FRA.

Next, we analyze the behavior of Sieve-SDP when applied to Shor's SDP relaxation.

Lemma 10 *Let $F = P \cap \{0,1\}^n$, where $P = \{x \in \mathbb{R}^n \mid Ax \leq b\}$, and let $L \cap \mathbb{S}^{n+1}_+$ denote Shor's relaxation for F, where L is defined in (3). Under Assumption 1, Sieve-SDP does not perform any reduction when applied to $L \cap \mathbb{S}^{n+1}_+$. In particular, $r_S = n+1$.*

Proof We examine all data matrices in Shor's relaxation (3) to determine whether Sieve-SDP can identify any exposing vectors. Note that a matrix can

be permuted into the form (27) only if it is positive semidefinite. The data matrices corresponding to the linear constraints $\mathcal{A}(Y) \leq \mathbf{0}$ are

$$A_i := \begin{bmatrix} -b_i & \frac{1}{2}a_i^T \\ \frac{1}{2}a_i & \mathbf{O} \end{bmatrix} \text{ for } i = 1, \ldots, m. \tag{28}$$

For any A_i to be positive semidefinite, it must be that $a_i = 0$. However, since $P \neq \emptyset$ by Assumption 1, this implies $b_i \geq 0$. Consequently, $A_i \notin \mathbb{S}_+^{n+1} \setminus \{\mathbf{O}\}$, so no matrix in (28) yields an exposing vector.

Next, consider the data matrices associated with the arrow constraint $\text{arrow}(Y) = e_0$:

$$\begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{O} \end{bmatrix} \text{ and } \begin{bmatrix} 0 & -\frac{1}{2}e_i^T \\ -\frac{1}{2}e_i & e_i e_i^T \end{bmatrix} \text{ for } i = 1, \ldots, n. \tag{29}$$

The first matrix is positive semidefinite, but its corresponding right-hand side is 1, and Sieve-SDP requires a right-hand side of zero to eliminate rows and columns. The remaining matrices in (29) are not positive semidefinite.

Therefore, none of the constraints satisfy the criteria for exposing vectors under Sieve-SDP, and no reduction occurs. It follows that $r_S = n + 1$.

We can extend Sieve-SDP as follows to capture more exposing vectors. In practice, Shor's SDP relaxation is often strengthened by incorporating valid equality constraints derived from the original problem. Specifically, for an equality constraint $a_i^T x = b_i$ ($a_i \neq \mathbf{0}$) in the original problem, one may add the valid equality constraint $\langle A_i, Y \rangle = 0$, where

$$A_i := \begin{bmatrix} -b_i \\ a_i \end{bmatrix} \begin{bmatrix} -b_i \\ a_i \end{bmatrix}^T.$$

As $A_i \neq \mathbf{O}$ is positive semidefinite, it is an exposing vector. Thus, by extending Sieve-SDP to allow D in (27) to be positive semidefinite, the exposing vector W^i can be detected by the modified algorithm. One can add up all exposing vectors identified in this way to construct an exposing vector with higher rank. Investigating the effectiveness and computational performance of this extension is an interesting direction for future research.

5 Numerical Experiments

In this section, we evaluate the performance of affine FR and compare it with partial FR. We demonstrate the effectiveness of affine FR in two main aspects. First, we examine the reduction in the size of the matrix variable and the time required for preprocessing. Second, we compare the total computation time for solving Shor's SDP relaxation with and without preprocessing via affine FR and partial FR.

All experiments were conducted on a MacBook Pro equipped with an Apple M2 Max processor, 128 GB of RAM, and macOS. We consider mixed-binary

linear programming instances from the MIPLIB 2017 [14], specifically those under the "collection" category with at most $10,000$ variables. The instances are read into MATLAB R2023b using Gurobi (version 10.0.1) [18]. We include only instances containing binary variables, continuous variables, or integer variables bounded between zero and one. Additionally, we exclude seven instances due to numerical issues[6] encountered when computing the affine hull of P.

The final dataset consists of 332 mixed-binary linear programming instances, representing a broad spectrum of application areas and covering a representative range of problem sizes relevant to SDP-based optimization methods. Larger instances typically become intractable without exploiting additional problem structure and cannot be solved within a reasonable timeframe. The MIPLIB 2017 instances are available at `https://miplib.zib.de/`. Our source code, including the list of tested instances, can be found at `https://github.com/haohu-code/affineFR_code`.

We use Gurobi to solve the LP problems that arise in affine FR and partial FR. For solving Shor's relaxation and its facially reduced formulations, we use Mosek (version 10.1) [3] in conjunction with YALMIP [26]. For affine FR, we compute the affine hull by solving the LP in (16). For partial FR, we implement the algorithm described in [33] to identify maximum-rank exposing vectors from $SEP(L \cap \mathcal{D}^*)$ and $SEP(L \cap \mathcal{DD}^*)$.[7] We impose a 5-minute time limit for all LPs used in preprocessing. If the solver fails to return a solution within this time limit, we classify the instance as yielding no reduction.

The mixed-binary linear programming instances from [14] take the following form:

$$\begin{aligned} \min\ & c^T x \\ \text{subject to } & Ax = b \\ & Bx \leq d \\ & l \leq x \leq u \\ & x_i \in \{0,1\} \quad \text{for } i \in \mathcal{B}, \end{aligned} \tag{30}$$

where $x \in \mathbb{R}^n$ is the decision variable. The linear system $Ax = b$ and $Bx \leq d$ represent the equality and inequality constraints, respectively. The vectors l and u specify the lower and upper bounds for x, respectively. If $l_i = -\infty$, then x_i has no lower bound. Similarly, $u_i = \infty$ means x_i has no upper bound. The subset $\mathcal{B} \subseteq \{1, \ldots, n\}$ indexes the binary variables.

[6] These instances are ill-conditioned, leading to numerical instability when solving (16). The excluded instances are `control20-5-10-5`, `control30-5-10-4`, `app1-1`, `app2-1`, `app2-2`, `lrn`, and `rentacar`.

[7] The original code in [33] targets generic SDP relaxations. For the cone of diagonally dominant matrices $\mathcal{DD}$, a direct implementation introduces $\mathcal{O}(n^2)$ additional constraints. By exploiting the arrowhead structure in Shor's relaxation, our customized implementation reduces this to $\mathcal{O}(n)$ constraints.

Shor's SDP relaxation for (30) is given by

$$\begin{aligned} \min\ & c^T x \\ \text{subject to } & Ax = b \\ & Bx \le d \\ & l \le x \le u \\ & x_i = X_{ii} \text{ for } i \in \mathcal{B} \\ & Y := \begin{bmatrix} 1 & x^T \\ x & X \end{bmatrix} \in \mathbb{S}_+^{n+1}. \end{aligned} \tag{31}$$

The order of the matrix variable Y is $n+1$. Applying an FRA, we construct a facial range vector $V \in \mathbb{R}^{(n+1)\times r}$ with $r \le n+1$, and replace Y with a smaller matrix variable $R \in \mathbb{S}_+^r$ via the substitution $Y = VRV^T$ (see (10)). In practice, problems with thousands of variables are common, and Shor's SDP relaxation often becomes computationally intractable due to memory constraints. As shown in the results below, the size of the matrix variable can be significantly reduced through facial reduction.

We emphasize that affine FR is independent of the specific SDP relaxation used to approximate the original mixed-binary linear program. That is, it can reduce the matrix dimension for any SDP relaxation. For example, strengthening Shor's SDP relaxation by imposing the nonnegativity constraint $Y \ge 0$ yields the so-called *doubly nonnegative relaxation*, which often provides tighter bounds at the cost of increased computational burden. Affine FR applies equally well to this formulation and yields reduction in matrix size.

5.1 Reduction in matrix size

In this section, we demonstrate the effectiveness and efficiency of affine FR in reducing the matrix size of SDP relaxations of mixed-binary linear programs. An overview of the results is presented in Tables 3 and 4. Detailed computational results are included in the Appendix.

Recall that n denotes the number of variables in a problem instance. The original SDP relaxation involves a matrix variable of order $n+1$. Table 3 summarizes the performance of three FRAs applied to the full set of 332 instances. Two metrics are used for comparison:

- "# Reduced Instances": Let r be the order of the matrix variable after preprocessing. A reduction is considered successful if $r < n+1$. The total number of instances that are successfully reduced by each method is listed in the second column.
- "Average Ratio": The ratio $\frac{r}{n+1}$, which lies in $(0, 1]$, indicates the relative size of the reduced matrix. A smaller ratio indicates a greater reduction. The final column reports the average of this ratio over all instances for each method.

As shown in Table 3, affine FR outperforms partial FR in both the number of reduced instances and the extent of matrix size reduction. It successfully

reduces a significantly larger portion of the test set, with deeper reduction in matrix size.

Method	# Reduced Instances	Average Ratio
Affine FR	265	0.8561
Partial FR ($\mathcal{D}^*$)	39	0.9889
Partial FR ($\mathcal{DD}^*$)	49	0.9846

Table 3 Summary of matrix size reduction across 332 mixed-binary instances

Table 4 reports the average preprocessing time for each FRA, grouped by problem size (i.e., the number of variables n). All three methods are LP-based and have comparable theoretical complexity. For each method, the reported time includes both the computation of an exposing vector and the construction of the corresponding facial range vector used to reformulate the SDP. In our experiments, we observe that the time spent on building the facial range vector dominates the time spent on computing the exposing vector. Note that if a method does not identify a nontrivial exposing vector, then the step of facial range vector computation is skipped. Since affine FR finds such vectors in significantly more instances, its average preprocessing time is expected to be longer. However, when restricted to instances where all three methods identify nontrivial exposing vectors, affine FR is evidently faster on average than the partial FR methods.

These results demonstrate that all three FRAs are computationally efficient and practical as preprocessing methods.

The range of n	Affine FR	Partial FR ($\mathcal{D}^*$)	Partial FR ($\mathcal{DD}^*$)
$[0, 2000]$	0.29	0.30	0.30
$(2000, 4000]$	1.88	2.60	2.60
$(4000, 6000]$	7.38	12.25	12.25
$(6000, 8000]$	10.12	13.10	13.10
$(8000, 10000]$	28.11	26.65	26.65

Table 4 Average preprocessing time (in seconds) by method and problem size

5.2 Reduction in computation time

In this section, we evaluate the impact of preprocessing on the time required to solve Shor's relaxation. We compare the performance of affine FR and partial FR with $\mathcal{DD}^*$. Partial FR with $\mathcal{D}^*$ is excluded from this comparison because it is generally weaker than partial FR with $\mathcal{DD}^*$ but takes similar computation time. Due to memory limitations, We restrict our experiments to instances with up to 350 variables.

Table 5 presents the computational results. Each column in the table provides the following information:

– "Instance": The name of the test problem.

- "Matrix size": The order of the matrix variable in the original or facially reduced Shor's relaxation.
- "Optimal value": The optimal value of the original mixed-binary program, as reported in MIPLIB 2017.
- "Lower bound (solver status)": The optimal value obtained from the SDP relaxation under each setting. If the solver encounters numerical issues, an asterisk is added in parentheses.
- "Prep. time": Time (in seconds) required for preprocessing, i.e., for computing an exposing vector and the corresponding facial range vector.
- "SDP time": Time (in seconds) used for constructing the facially reduced problem (if applicable) and solving it using Mosek.
- "Time Ratio": A measure of efficiency, defined as:

$$\frac{\text{(SDP time + Prep. time) using FRA}}{\text{SDP time without FR}}.$$

The results clearly demonstrate that affine FR can significantly reduce the computation time for solving Shor's relaxation. In many instances, preprocessing with affine FR results in substantial speedups. However, there are a few exceptions where preprocessing increases total computation time. This typically occurs when affine FR yields only a minor reduction — often reducing the matrix size by just one. In such cases, the overhead of building a facially reduced formulation may outweigh the computational benefit, particularly if sparsity or other problem structures are not preserved during reduction. For these cases, solving the original Shor's relaxation may be more efficient if the reduction ratio falls below a certain threshold. For further discussion on the importance of sparsity preservation, we refer the reader to [33].

5.3 Improvement in numerical stability

In our numerical experiment, we observe that the optimal values of Shor's SDP relaxation and its facially reduced counterpart can differ. This section discusses the reasons for such discrepancies, focusing on numerical stability and the role of facial reduction.

When an SDP problem fails to satisfy Slater's condition, solvers may return unreliable results due to the lack of strict feasibility. In such cases, facial reduction can help by identifying a smaller face of the positive semidefinite cone where Slater's condition is restored. If the failure of Slater's condition is the sole source of numerical instability, then facial reduction theoretically improves numerical behavior, and the lower bound obtained from the facially reduced problem is typically more reliable.

To illustrate this, consider the SDP with feasible region $L \cap \mathbb{S}^n_+$, where

$$L := \{X \in \mathbb{S}^n \mid X_{11} = 1, X_{22} = 0, X_{i+1,i+1} = X_{1i} \ \text{ for } i = 2, \ldots, n-1\}.$$

The set $L \cap \mathbb{S}^n_+$ contains a unique feasible solution: a matrix with a one in the top-left entry and zeros elsewhere. However, this set fails to satisfy Slater's

Instance	Optimal value	Without FR			Affine FR					Partial FR ($\mathcal{DD}^*$)				
		Matrix size	Lower bound (solver status)	SDP time	Matrix size	Lower bound (solver status)	Prep. time	SDP time	Time ratio	Matrix size	Lower bound (solver status)	Prep. time	SDP time	Time ratio
markshare-4-0	1.000	35	0.000	0.04	31	0.000	0.011	0.03	1.00	35	0.000	0.005	0.04	1.12
markshare-5-0	1.000	46	0.000	0.10	41	0.000	0.007	0.08	0.85	46	0.000	0.005	0.10	1.05
g503inf	infeasible	49	195.924	0.18	40	195.924	0.010	0.12	0.74	49	195.924	0.007	0.18	1.04
gr4x6	202.350	49	185.550	0.22	40	185.550	0.006	0.13	0.60	49	185.550	0.005	0.22	1.02
markshare1	1.000	63	0.000	0.32	51	0.000	0.008	0.19	0.63	63	0.000	0.006	0.32	1.02
pb-market-split8-70-4	0.000	72	0.000	0.73	63	0.000	0.010	0.36	0.51	71	0.000	0.010	0.75	1.03
markshare2	1.000	75	-0.000	0.89	61	0.000	0.008	0.34	0.39	75	-0.000	0.007	0.89	1.01
pk1	11.000	87	0.000	2.11	72	0.000	0.009	0.80	0.38	87	0.000	0.006	2.11	1.00
b-ball	-1.500	101	-1.818	2.29	82	-1.818	0.007	1.42	0.62	101	-1.818	0.008	2.29	1.00
p2m2p1m1p0n100	infeasible	102	80424.000	4.89	101	80423.986	0.019	4.85	1.00	102	80424.000	0.010	4.89	1.00
misc05inf	infeasible	137	2948.950 (*)	42.49	102	2949.887 (*)	0.026	15.16	0.36	133	2949.632 (*)	0.025	30.41	0.72
v150d30-2hopcds	41.000	151	27.617	28.91	149	27.617	0.094	29.23	1.01	149	27.617	0.055	29.30	1.02
assign1-5-8	212.000	157	183.363	29.33	127	183.363	0.030	12.34	0.42	157	183.363	0.015	29.33	1.00
neos-5192052-neckar	-11670000.000	181	-11859802.231	254.45	161	-11980000.000	0.022	176.30	0.69	181	-11859802.231	0.012	254.45	1.00
neos-3135526-osun	infeasible	193	0.000 (*)	206.62	161	175.096 (*)	0.059	186.45	0.90	193	0.000 (*)	0.029	206.62	1.00
neos-5140963-mincio	14393.000	197	8876.999	137.87	170	8877.000	0.032	65.52	0.48	197	8876.999	0.015	137.87	1.00
p0201	7615.000	202	6875.000	72.04	146	6875.000	0.043	37.25	0.52	202	6875.000	0.016	72.04	1.00
mad	0.027	221	0.000	99.57	191	0.000	0.024	73.02	0.73	221	0.000	0.019	99.57	1.00
prod1	-56.000	251	-100.000	391.02	243	-100.000	0.038	303.83	0.78	251	-100.000	0.023	391.02	1.00
misc07	2810.000	261	1422.584 (*)	857.87	208	1446.909	0.043	473.20	0.55	240	1415.582	0.023	587.10	0.68
prod2	-62.000	302	-100.000	1370.95	291	-100.000	0.045	832.74	0.61	302	-100.000	0.032	1370.95	1.00
supportcase14	288.000	305	272.000	1077.33	179	272.000	0.059	82.31	0.08	301	272.000	0.063	931.14	0.86
supportcase16	288.000	320	272.000	1701.96	193	272.000	0.045	142.43	0.08	316	272.000	0.090	1318.07	0.77
probportfolio	16.734	321	5.000	1424.44	320	5.000	0.078	1667.25	1.17	321	5.000	0.034	1424.44	1.00
glass4	1200012599.972	323	800002399.811	2637.57	286	800002396.579	0.052	1232.16	0.47	323	800002399.811	0.036	2637.57	1.00
control30-3-2-3	-17.257	333	-9450.426 (*)	3712.46	179	-178.921 (*)	0.071	104.71	0.03	333	-9450.426 (*)	0.071	3712.46	1.00
ran13x13	3252.000	339	2691.439	2572.25	314	2691.439	0.054	1873.19	0.73	339	2691.439	0.042	2572.25	1.00
pigeon-08	-7000.000	345	-8000.000	2193.84	305	-8000.000	0.057	880.15	0.40	345	-8000.000	0.048	2193.84	1.00

Table 5 Performance comparison of solving Shor's SDP relaxation with no preprocessing, affine FR, and partial FR ($\mathcal{DD}^*$)

condition because it lies entirely on the boundary of the cone. As a result, even state-of-the-art solvers may produce inaccurate results [34,38,42]. For instance, when $n = 5$, SDPT3 (via `cvx`) reports the top-left entry as 0.94, as documented in [38]. Similar discrepancies can arise with other solvers.

In contrast, this numerical issue does not arise in the facially reduced formulation for $L \cap \mathbb{S}^n_+$. The facially reduced formulation eliminates the problematic variables and operates over a smaller subspace where Slater's condition is satisfied. Solvers applied to the reduced problem typically produce much more accurate solutions.

However, it is important to note that facial reduction is not a universal remedy. Numerical issues may also arise from poor problem scaling, modeling errors, or ill-conditioned data. For example, if a problem includes a constraint such as $x_1 \leq 10^{-12}$, it may be unclear whether x_1 should be considered truly zero. In such cases, even after facial reduction, solvers may still return inaccurate solutions unless additional structure or domain knowledge is leveraged.

In summary, while facial reduction often improves numerical stability, especially when Slater's condition is the underlying issue, it does not guarantee correctness in all situations. Careful modeling and interpretation remain essential for reliable optimization.

6 Conclusion

In this paper, we proposed a novel FRA, affine FR, for SDP relaxations of CO problems. SDP relaxations are powerful tools for solving challenging CO problems, as they often yield extremely tight bounds. However, a major bottleneck in their application is the frequent failure of regularity conditions such as Slater's condition, which limits the scalability and reliability of SDP-based methods. Addressing this issue is crucial for extending the reach of SDP techniques to broader classes and larger instances of CO problems, and thus represent an important direction in optimization research.

Affine FR is a fully automated preprocessing algorithm that exploits the inherent structures present in CO problems, leading to enhanced performance. It serves as a valuable addition to the existing approaches for CO problems. We provide a theoretical analysis to demonstrate the differences between affine FR and existing methods. Furthermore, based on our experiments presented in Section 5, we observe that affine FR demonstrates remarkable potential in preprocessing SDP relaxations of CO problems. Our findings underscore the practical effectiveness and suitability of affine FR.

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A Detailed Results

Tables 6–9 present the detailed computational results that are summarized in Tables 3 and 4. We tested 332 instances of mixed-binary linear programming from the MIPLIP 2017 dataset, each with up to 10,000 variables. Only instances that exhibit a positive reduction in matrix size from at least one of the three FRAs are included.

The first column lists the name of each instance. The second column reports the original matrix size, $n + 1$, in Shor's relaxation. The columns labeled r_A, r_P, and r_P^+ represent the matrix sizes after applying affine FR, partial FR with $\mathcal{D}^*$, and partial FR with $\mathcal{DD}^*$, respectively. The corresponding reduction ratio, defined as the reduced matrix size divided by the original matrix size, e.g.,, $\frac{r_A}{n+1}$ for affine FR, are also reported. A smaller ratio indicates a more substantial reduction. Finally, the column "time" gives the computation time in seconds required for computing an exposing vector.

An high-level discussion of these results can be found in Section 5.1. A few additional remarks on Tables 6–9 are provided below:

- The instance `neos-807639` achieves the most significant reduction, with the matrix size reduced to just 17.6% of its original value.
- Some instances exhibit only minimal reductions. For instance, Shor's relaxation for `rmatr100-p10` originally has a matrix of size 7360, which is reduced to 7539 by affine FR. The ratio $\frac{7359}{7360}$ rounds to 1.000 when displayed to three decimal places in the table.
- The computation time of the three FRAs is typically negligible compared to the time required to solve the corresponding facially reduced SDP.
- A time limit of 5 minutes is imposed for all preprocessing runs. For the instance `neos-3083784-nive`, partial FR with both $\mathcal{D}^*$ and $\mathcal{DD}^*$ reached this time limit.

Instance	Matrix Size	Affine FR			Partial FR ($\mathcal{D}^*$)			Partial FR ($\mathcal{DD}^*$)		
	$n+1$	r_A	ratio	time	r_P	ratio	time	$r_P^{\perp}$	ratio	time
markshare-4-0	35	31	0.886	0.005	35	1.000	0.005	35	1.000	0.005
markshare-5-0	46	41	0.891	0.006	46	1.000	0.005	46	1.000	0.005
g503inf	49	40	0.816	0.007	49	1.000	0.006	49	1.000	0.006
gr4x6	49	40	0.816	0.005	49	1.000	0.006	49	1.000	0.005
markshare1	63	51	0.810	0.006	63	1.000	0.005	63	1.000	0.005
pb-market-split8-70-4	72	63	0.875	0.007	71	0.986	0.008	71	0.986	0.008
markshare2	75	61	0.813	0.008	75	1.000	0.007	75	1.000	0.006
pk1	87	72	0.828	0.009	87	1.000	0.006	87	1.000	0.006
b-ball	101	82	0.812	0.011	101	1.000	0.008	101	1.000	0.008
p2m2p1m1p0n100	102	101	0.990	0.034	102	1.000	0.012	102	1.000	0.010
misc05inf	137	102	0.745	0.026	135	0.985	0.015	133	0.971	0.028
v150d30-2hopcds	151	149	0.987	0.101	151	1.000	0.053	149	0.987	0.062
assign1-5-8	157	127	0.809	0.025	157	1.000	0.012	157	1.000	0.013
neos-5192052-neckar	181	161	0.890	0.026	181	1.000	0.012	181	1.000	0.012
neos-3135526-osun	193	161	0.834	0.058	193	1.000	0.024	193	1.000	0.027
neos-5140963-mincio	197	170	0.863	0.032	197	1.000	0.014	197	1.000	0.014
p0201	202	146	0.723	0.037	202	1.000	0.015	202	1.000	0.015
mad	221	191	0.864	0.037	221	1.000	0.017	221	1.000	0.018
prod1	251	243	0.968	0.046	251	1.000	0.023	251	1.000	0.024
misc07	261	208	0.797	0.048	240	0.920	0.022	240	0.920	0.024
prod2	302	291	0.964	0.059	302	1.000	0.029	302	1.000	0.031
supportcase14	305	179	0.587	0.061	304	0.997	0.032	301	0.987	0.078
supportcase16	320	193	0.603	0.063	319	0.997	0.036	316	0.988	0.090
probportfolio	321	320	0.997	0.081	321	1.000	0.030	321	1.000	0.032
glass4	323	286	0.885	0.061	323	1.000	0.029	323	1.000	0.035
control30-3-2-3	333	179	0.538	0.067	333	1.000	0.076	333	1.000	0.079
ran13x13	339	314	0.926	0.058	339	1.000	0.030	339	1.000	0.031
pigeon-08	345	305	0.884	0.063	345	1.000	0.030	345	1.000	0.032
23588	369	232	0.629	0.080	369	1.000	0.066	369	1.000	0.071
nsa	389	352	0.905	0.085	389	1.000	0.036	389	1.000	0.037
beavma	391	303	0.775	0.083	391	1.000	0.037	391	1.000	0.106
neos-3611689-kaihu	422	421	0.998	0.090	422	1.000	0.039	422	1.000	0.043
22433	430	232	0.540	0.149	430	1.000	0.086	430	1.000	0.112
neos-3610040-iskar	431	430	0.998	0.088	431	1.000	0.040	431	1.000	0.050
rlp1	462	364	0.788	0.102	372	0.805	0.048	372	0.805	0.050
neos-3611447-jijia	473	472	0.998	0.101	473	1.000	0.053	473	1.000	0.051
k16x240b	481	466	0.969	0.110	481	1.000	0.050	481	1.000	0.056
nexp-50-20-1-1	491	462	0.941	0.113	491	1.000	0.144	489	0.996	0.152
pigeon-10	491	441	0.898	0.117	491	1.000	0.052	491	1.000	0.056
neos-2629914-sudost	497	466	0.938	0.197	497	1.000	0.103	497	1.000	0.106
ran12x21	505	473	0.937	0.121	505	1.000	0.058	505	1.000	0.057
ran14x18-disj-8	505	474	0.939	0.121	505	1.000	0.060	505	1.000	0.163
bienst1	506	379	0.749	0.126	506	1.000	0.156	506	1.000	0.059
bienst2	506	379	0.749	0.124	506	1.000	0.058	506	1.000	0.056
newdano	506	379	0.749	0.125	506	1.000	0.160	506	1.000	0.063
milo-v13-4-3d-3-0	517	181	0.350	0.146	517	1.000	0.069	517	1.000	0.069
danoint	522	379	0.726	0.141	522	1.000	0.205	522	1.000	0.086
neos-480878	535	499	0.933	0.156	535	1.000	0.088	535	1.000	0.086
neos17	536	511	0.953	0.141	536	1.000	0.071	536	1.000	0.085
dcmulti	549	471	0.858	0.152	549	1.000	0.081	549	1.000	0.082
assign1-10-4	573	512	0.894	0.187	573	1.000	0.089	573	1.000	0.259
neos-3072252-nete	577	434	0.752	0.171	577	1.000	0.087	577	1.000	0.086
sp150x300d	601	452	0.752	0.186	601	1.000	0.089	601	1.000	0.093
rd-rplusc-21	623	485	0.778	0.505	623	1.000	0.305	623	1.000	0.311
milo-v13-4-3d-4-0	689	241	0.350	0.252	689	1.000	0.121	689	1.000	0.114
r50x360	721	672	0.932	0.241	721	1.000	0.345	721	1.000	0.133
pigeon-13	755	690	0.914	0.261	755	1.000	0.145	755	1.000	0.141
opt1217	770	722	0.938	0.264	770	1.000	0.140	770	1.000	0.144
fastxgemm-n2r6s0t2	785	641	0.817	0.286	785	1.000	0.151	785	1.000	0.151
bg512142	793	543	0.685	0.287	793	1.000	0.174	783	0.987	0.415
neos-3610051-istra	806	805	0.999	0.266	806	1.000	0.162	806	1.000	0.151
qiu	841	709	0.843	0.295	841	1.000	0.170	841	1.000	0.184
aflow30a	843	785	0.931	0.308	843	1.000	0.170	843	1.000	0.183
neos-3610173-itata	845	844	0.999	0.284	845	1.000	0.164	845	1.000	0.176
neos-911970	889	854	0.961	0.314	889	1.000	0.178	889	1.000	0.186
supportcase20	904	601	0.665	0.387	904	1.000	0.246	904	1.000	0.254
fastxgemm-n2r7s4t1	905	737	0.814	0.341	905	1.000	0.196	905	1.000	0.199

Table 6 Detailed matrix size reduction across 332 mixed-binary instances

Instance	Matrix Size	Affine FR			Partial FR ($\mathcal{D}^*$)			Partial FR ($\mathcal{DD}^*$)		
	$n+1$	r_A	ratio	time	r_P	ratio	time	r_P^+	ratio	time
ponderthis0517-inf	976	899	0.921	0.370	976	1.000	0.256	976	1.000	0.603
exp-1-500-5-5	991	741	0.748	0.376	991	1.000	0.228	991	1.000	0.233
neos-4333596-skien	1006	753	0.749	0.399	1006	1.000	0.248	1006	1.000	0.245
supportcase30	1025	1021	0.996	0.412	1025	1.000	0.240	1025	1.000	0.247
neos-807639	1031	181	0.176	0.389	1031	1.000	0.247	1031	1.000	0.256
pigeon-16	1073	993	0.925	0.386	1073	1.000	0.261	1073	1.000	0.263
tr12-30	1081	721	0.667	0.406	1081	1.000	0.276	1081	1.000	0.283
berlin-5-8-0	1084	989	0.912	0.412	1084	1.000	0.292	1084	1.000	0.309
neos-631517	1091	924	0.847	0.393	1091	1.000	0.261	1091	1.000	0.271
neos-3118745-obra	1132	1131	0.999	0.429	1132	1.000	0.272	1132	1.000	0.276
neos-1396125	1162	781	0.672	0.458	1162	1.000	0.760	1162	1.000	0.751
gmu-35-40	1206	1201	0.996	0.463	1206	1.000	0.330	1206	1.000	0.345
nexp-50-20-4-2	1226	1205	0.983	0.491	1226	1.000	0.336	1226	1.000	0.328
fiber	1299	947	0.729	0.595	1288	0.992	0.436	1287	0.991	0.961
hgms-det	1323	784	0.593	0.672	1323	1.000	1.023	1300	0.983	0.990
acc-tight5	1340	442	0.330	1.005	1281	0.956	1.466	1281	0.956	1.696
khb05250	1351	1225	0.907	0.609	1351	1.000	0.390	1351	1.000	1.129
seymour	1373	1256	0.915	0.753	1373	1.000	0.412	1256	0.915	1.136
seymour1	1373	1256	0.915	0.773	1373	1.000	0.488	1294	0.942	1.132
neos-860300	1386	1216	0.877	1.118	1234	0.890	1.226	1234	0.890	0.537
neos-4650160-yukon	1413	941	0.666	0.801	1413	1.000	0.456	1413	1.000	0.459
bppc4-08	1457	1436	0.986	0.815	1457	1.000	0.568	1457	1.000	0.528
neos-3627168-kasai	1463	998	0.682	0.863	1463	1.000	0.504	1463	1.000	0.536
g200x740	1481	1282	0.866	0.820	1481	1.000	0.527	1481	1.000	0.543
neos-4954672-berkel	1534	1009	0.658	0.900	1534	1.000	0.540	1534	1.000	0.582
csched008	1537	1236	0.804	1.036	1537	1.000	1.594	1537	1.000	0.612
pigeon-20	1581	1481	0.937	0.906	1581	1.000	0.608	1581	1.000	0.660
acc-tight2	1621	1344	0.829	0.957	1621	1.000	1.882	1621	1.000	0.980
acc-tight4	1621	661	0.408	0.999	1621	1.000	0.950	1621	1.000	0.945
ns1830653	1630	562	0.345	1.056	884	0.542	1.878	866	0.531	1.972
eva1aprime5x5opt	1713	1409	0.823	1.431	1713	1.000	0.930	1713	1.000	0.868
bc	1752	967	0.552	1.282	1752	1.000	1.114	1752	1.000	1.172
bc1	1752	967	0.552	1.217	1752	1.000	1.096	1752	1.000	1.078
csched007	1759	1458	0.829	1.177	1759	1.000	2.096	1759	1.000	0.838
csched010	1759	1458	0.829	1.881	1759	1.000	1.850	1759	1.000	1.621
railway-8-1-0	1797	1592	0.886	0.942	1797	1.000	0.813	1797	1.000	0.823
aligninq	1832	1831	0.999	0.100	1832	1.000	0.802	1832	1.000	0.822
gmu-35-50	1920	1915	0.997	1.143	1920	1.000	0.848	1920	1.000	0.898
mtest4ma	1951	1852	0.949	1.233	1951	1.000	1.086	1951	1.000	0.942
neos-5182409-nasivi	2006	1742	0.868	1.626	2006	1.000	1.187	2006	1.000	1.043
10teams	2026	1459	0.720	1.934	1826	0.901	2.902	1826	0.901	3.065
dg012142	2081	1401	0.673	1.653	2081	1.000	1.115	2041	0.981	2.446
neos2	2102	2059	0.980	1.521	2102	1.000	1.084	2102	1.000	1.090
neos-585467	2117	855	0.404	1.508	2117	1.000	1.189	2117	1.000	1.208
graph20-20-1rand	2184	1925	0.881	1.408	2147	0.983	1.248	2147	0.983	1.241
neos-4477313-unzha	2194	2052	0.935	1.513	2194	1.000	1.149	2194	1.000	1.140
neos-2328163-agri	2237	2160	0.966	1.489	2237	1.000	2.366	2237	1.000	2.361
uct-subprob	2257	1356	0.601	0.986	2257	1.000	1.256	2257	1.000	1.585
binkar10-1	2299	1283	0.558	1.118	2299	1.000	1.261	2299	1.000	1.298
chromaticindex32-8	2305	1730	0.751	1.056	2305	1.000	1.256	2305	1.000	1.333
usAbbrv-8-25-70	2313	2104	0.910	1.371	2313	1.000	1.362	2313	1.000	1.262
p200x1188c	2377	2178	0.916	1.091	2377	1.000	1.328	2377	1.000	1.366
tbfp-bigm	2407	2377	0.988	1.276	2407	1.000	1.520	2407	1.000	1.537
beasleyC1	2501	2002	0.800	1.387	2501	1.000	1.450	2501	1.000	1.493
beasleyC2	2501	2002	0.800	1.159	2501	1.000	1.384	2501	1.000	1.419
beasleyC3	2501	2000	0.800	1.414	2501	1.000	1.421	2501	1.000	1.419
neos-859770	2505	2304	0.920	2.350	2321	0.927	1.656	2321	0.927	1.730
neos-585192	2598	1054	0.406	1.283	2598	1.000	1.754	2598	1.000	1.834
mod010	2656	2430	0.915	1.675	2573	0.969	1.714	2573	0.969	1.873
neos-1330346	2665	2341	0.878	1.415	2665	1.000	1.681	2665	1.000	1.694
milo-v12-6-r2-40-1	2689	1681	0.625	1.540	2689	1.000	2.041	2689	1.000	1.957
circ10-3	2701	2251	0.833	1.725	2701	1.000	2.145	2701	1.000	2.232
pg	2701	2601	0.963	1.426	2701	1.000	1.613	2701	1.000	1.661
ns2071214	2721	1560	0.573	1.431	2657	0.976	1.934	2657	0.976	1.937
aflow40b	2729	2651	0.971	1.622	2729	1.000	1.764	2729	1.000	1.794
neos-4333464-siret	2732	2651	0.970	1.323	2732	1.000	1.762	2732	1.000	1.808
neos-1337307	2841	2715	0.956	1.504	2841	1.000	2.094	2841	1.000	2.255

Table 7 Detailed matrix size reduction across 332 mixed-binary instances

Instance	Matrix Size	Affine FR			Partial FR ($\mathcal{D}^*$)			Partial FR ($\mathcal{DD}^*$)		
	$n+1$	r_A	ratio	time	r_P	ratio	time	r_P^+	ratio	time
neos-503737	2851	2461	0.863	1.546	2601	0.912	1.833	2601	0.912	1.871
ns1208400	2884	2261	0.784	1.648	2600	0.902	3.010	2600	0.902	3.034
neos-5261882-treska	2901	2716	0.936	1.818	2901	1.000	2.157	2901	1.000	2.238
neos-3402294-bobin	2905	2697	0.928	4.580	2905	1.000	3.553	2905	1.000	3.556
neos-3402454-bohle	2905	2697	0.928	16.968	2905	1.000	9.482	2905	1.000	9.740
supportcase2	2913	2529	0.868	11.955	2913	1.000	9.078	2913	1.000	8.689
s1234	2946	2865	0.973	1.569	2946	1.000	1.946	2946	1.000	2.029
lotsize	2986	2386	0.799	1.560	2986	1.000	1.936	2986	1.000	2.022
neos-691058	3007	2505	0.833	1.657	2799	0.931	3.886	2799	0.931	4.065
neos-933562	3033	957	0.316	1.710	3033	1.000	4.389	3033	1.000	4.674
mc11	3041	2642	0.869	2.161	3041	1.000	2.135	3041	1.000	2.163
mc7	3041	2642	0.869	1.800	3041	1.000	2.082	3041	1.000	2.076
mc8	3041	2642	0.869	1.969	3041	1.000	2.106	3041	1.000	2.058
app3	3081	1553	0.504	2.113	3081	1.000	2.092	3081	1.000	2.299
neos-5107597-kakapo	3115	3046	0.978	2.036	3115	1.000	2.292	3115	1.000	2.397
neos-5115478-kaveri	3115	3046	0.978	1.938	3115	1.000	2.198	3115	1.000	2.280
supportcase4	3163	1113	0.352	2.044	2663	0.842	2.343	2139	0.676	4.513
neos22	3241	3215	0.992	1.999	3241	1.000	2.443	3241	1.000	2.553
neos-1112787	3281	3238	0.987	2.060	3281	1.000	2.793	3281	1.000	2.915
dfn-bwin-DBE	3286	3193	0.972	2.022	3286	1.000	2.592	3286	1.000	4.280
neos18	3313	964	0.291	2.057	3283	0.991	2.805	1019	0.308	4.954
neos-619167	3453	3313	0.959	2.129	3453	1.000	2.831	3453	1.000	2.796
neos-5078479-escaut	3472	2432	0.700	2.199	3472	1.000	3.095	3472	1.000	3.130
eva1aprime6x6opt	3515	2937	0.836	3.170	3515	1.000	3.916	3515	1.000	3.716
neos-3216931-puriri	3556	2992	0.841	2.797	3556	1.000	3.867	3556	1.000	6.790
bnatt400	3601	2014	0.559	2.476	2015	0.560	3.026	2015	0.560	3.027
neos-525149	3641	3522	0.967	2.969	3641	1.000	3.447	3641	1.000	3.546
a1c1s1	3649	2401	0.658	2.788	3649	1.000	3.039	3649	1.000	3.096
a2c1s1	3649	2401	0.658	2.450	3649	1.000	2.942	3649	1.000	3.050
hanoi5	3863	2976	0.770	3.412	3513	0.909	3.693	3503	0.907	6.464
b1c1s1	3873	2593	0.670	2.825	3873	1.000	3.432	3873	1.000	3.581
b2c1s1	3873	2593	0.670	3.000	3873	1.000	3.358	3873	1.000	3.497
neos-4954357-bednja	3886	3854	0.992	2.706	3886	1.000	3.632	3886	1.000	3.697
bppc6-06	3923	3902	0.995	2.997	3923	1.000	3.517	3923	1.000	3.609
fhnw-binpack4-77	3925	3527	0.899	6.234	3924	1.000	3.652	3528	0.899	6.247
neos-4387871-tavua	4005	3723	0.930	2.825	4005	1.000	3.839	4005	1.000	3.724
neos-4393408-tinui	4005	3723	0.930	2.821	4005	1.000	3.581	4005	1.000	3.645
set3-09	4020	2769	0.689	3.034	4020	1.000	4.160	4020	1.000	4.441
set3-10	4020	2769	0.689	2.912	4020	1.000	3.752	4020	1.000	4.004
set3-15	4020	2769	0.689	3.073	4020	1.000	3.978	4020	1.000	4.043
set3-16	4020	2769	0.689	3.235	4020	1.000	3.985	4020	1.000	4.158
set3-20	4020	2769	0.689	3.109	4020	1.000	3.925	4020	1.000	4.041
neos-1605061	4112	2681	0.652	196.900	3362	0.818	8.236	3362	0.818	8.276
neos-1112782	4141	4092	0.988	3.041	4141	1.000	4.385	4141	1.000	4.571
qap10	4151	2603	0.627	3.031	4151	1.000	9.099	4151	1.000	10.006
cost266-UUE	4162	2866	0.689	3.010	4162	1.000	6.756	4162	1.000	7.151
neos-5178119-nalagi	4168	4015	0.963	3.497	4168	1.000	4.338	4168	1.000	4.393
supportcase3	4192	2141	0.511	3.253	3678	0.877	4.392	3168	0.756	7.094
neos-3634244-kauru	4377	4178	0.955	4.614	4377	1.000	5.575	4377	1.000	5.462
milo-v12-6-r1-58-1	4441	3002	0.676	3.453	4441	1.000	4.627	4441	1.000	4.774
neos-1601936	4447	3001	0.675	3.755	3697	0.831	8.293	3697	0.831	8.228
nexp-150-20-1-5	4471	4413	0.987	3.489	4471	1.000	4.508	4471	1.000	4.799
bnatt500	4501	2530	0.562	0.403	2530	0.562	4.551	2530	0.562	4.704
neos-595904	4509	4357	0.966	3.557	4509	1.000	4.697	4509	1.000	4.954
eil33-2	4517	4485	0.993	3.948	4517	1.000	4.664	4517	1.000	5.804
peg-solitaire-a3	4553	3186	0.700	4.185	4552	1.000	6.043	4516	0.992	9.885
neos-831188	4613	3487	0.756	4.348	4613	1.000	5.083	4613	1.000	5.413
neos-3762025-ognon	4674	3049	0.652	4.178	4674	1.000	5.588	4674	1.000	5.472
neos-3660371-kurow	4723	2749	0.582	4.178	4723	1.000	5.365	4723	1.000	5.123
supportcase41	4726	4603	0.974	1.875	4686	0.992	6.380	4686	0.992	6.387
neos-3065804-namu	4742	3644	0.768	5.171	4742	1.000	5.546	4742	1.000	5.839
neos-3083784-nive	4780	3987	0.834	4.195	4780	1.000	303.729	4780	1.000	304.144
bppc6-02	4785	4764	0.996	4.487	4785	1.000	5.387	4785	1.000	5.488
misc04inf	4898	4315	0.881	4.774	4898	1.000	6.782	4898	1.000	6.566
h50x2450	4901	4803	0.980	4.475	4901	1.000	9.696	4901	1.000	5.823
neos-1420790	4927	2647	0.537	4.352	4927	1.000	5.586	4927	1.000	5.790
neos-5075914-elvire	5004	3768	0.753	4.442	5004	1.000	6.311	5004	1.000	23.432

Table 8 Detailed matrix size reduction across 332 mixed-binary instances

Instance	Matrix Size	Affine FR			Partial FR ($\mathcal{D}^*$)			Partial FR ($\mathcal{D}\mathcal{D}^*$)		
	$n+1$	r_A	ratio	time	r_P	ratio	time	r_P^+	ratio	time
neos-3116779-oban	5142	5141	1.000	5.673	5142	1.000	6.064	5142	1.000	6.173
fhnw-schedule-paira100	5151	5124	0.995	0.282	5124	0.995	6.295	5124	0.995	6.288
momentum1	5175	3088	0.597	5.633	5030	0.972	12.671	5028	0.972	12.228
neos-1324574	5257	4645	0.884	5.328	5257	1.000	6.514	5257	1.000	6.671
istanbul-no-cutoff	5283	5062	0.958	5.628	5283	1.000	6.941	5283	1.000	7.140
berlin	5305	5254	0.990	5.200	5305	1.000	6.612	5305	1.000	6.806
mkc	5326	5324	1.000	5.598	5326	1.000	6.539	5326	1.000	6.921
mkc1	5326	5324	1.000	5.233	5326	1.000	6.574	5326	1.000	6.694
var-smallemery-m6j6	5609	5608	1.000	3.119	5609	1.000	7.718	5609	1.000	8.069
milo-v12-6-r1-75-1	5699	3852	0.676	6.515	5699	1.000	8.248	5699	1.000	8.386
bley-xl1	5832	3695	0.634	12.626	5832	1.000	9.083	5818	0.998	15.290
neos-826650	5913	4197	0.710	6.772	5913	1.000	10.182	5913	1.000	9.950
p500x2988	5977	5478	0.917	6.895	5977	1.000	8.824	5977	1.000	8.740
p500x2988c	5977	5478	0.917	6.604	5977	1.000	8.353	5977	1.000	8.523
p500x2988d	5977	5478	0.917	6.772	5977	1.000	8.590	5977	1.000	8.556
cap6000	6001	5878	0.980	7.307	6001	1.000	8.720	6001	1.000	8.928
blp-ir98	6098	6029	0.989	8.251	6098	1.000	9.065	6098	1.000	9.332
neos-3695882-vesdre	6136	5927	0.966	8.148	6136	1.000	14.139	6136	1.000	13.820
neos-4960896-besbre	6150	6105	0.993	7.988	6150	1.000	9.097	6150	1.000	9.046
atm20-100	6481	4441	0.685	9.012	6481	1.000	10.439	6481	1.000	10.908
neos-3761878-oglio	6490	4118	0.635	8.886	6490	1.000	10.737	6490	1.000	11.144
neos-4382714-ruvuma	6563	3841	0.585	9.306	6563	1.000	20.043	6563	1.000	21.889
brasil	6613	6556	0.991	9.817	6613	1.000	10.141	6613	1.000	10.524
leo1	6732	3626	0.539	10.843	6732	1.000	10.647	6732	1.000	10.927
supportcase25	6770	3601	0.532	8.984	6770	1.000	12.107	6770	1.000	12.551
swath	6806	6303	0.926	10.191	6405	0.941	11.097	6405	0.941	10.968
swath1	6806	6303	0.926	10.122	6617	0.972	10.674	6617	0.972	10.957
swath2	6806	6303	0.926	9.436	6613	0.972	10.891	6613	0.972	10.755
swath3	6806	6303	0.926	9.648	6597	0.969	11.244	6597	0.969	10.648
roi2alpha3n4	6817	6815	1.000	13.715	6817	1.000	11.371	6817	1.000	11.288
neos-3218348-suir	7023	6903	0.983	11.031	7023	1.000	11.623	7023	1.000	12.434
supportcase22	7130	6490	0.910	14.498	7130	1.000	14.291	7130	1.000	15.434
air05	7196	5885	0.818	12.277	6310	0.877	24.212	6310	0.877	25.217
biella1	7329	6132	0.837	11.345	7329	1.000	12.910	7329	1.000	12.979
rmatr100-p10	7360	7359	1.000	11.136	7360	1.000	12.067	7360	1.000	12.605
maxgasflow	7438	5151	0.693	12.128	7438	1.000	13.216	7438	1.000	13.397
fhnw-binpack4-58	7551	6951	0.921	12.655	7551	1.000	13.370	6951	0.921	25.355
trento1	7688	6440	0.838	12.796	7688	1.000	13.914	7688	1.000	14.854
neos-4954340-beaury	7851	7778	0.991	21.285	7851	1.000	14.319	7851	1.000	14.636
cdma	7892	6583	0.834	20.515	7892	1.000	14.361	7892	1.000	14.838
neos-3581454-haast	8113	7477	0.922	21.964	8113	1.000	15.576	8113	1.000	16.576
neos-3209519-ruhr	8676	8001	0.922	25.166	8676	1.000	42.954	8676	1.000	20.711
neos-1067731	8780	5311	0.605	30.080	8780	1.000	21.127	8780	1.000	21.036
rmatr100-p5	8785	8784	1.000	24.831	8785	1.000	19.217	8785	1.000	19.149
neos6	8787	8564	0.975	28.355	8787	1.000	18.981	8787	1.000	19.073
neos-3703351-marne	8828	5322	0.603	27.358	8828	1.000	22.387	8828	1.000	21.782
air04	8905	6545	0.735	30.478	7343	0.825	54.287	7343	0.825	56.298
neos-983171	8966	8808	0.982	28.786	8966	1.000	20.466	8966	1.000	20.288
neos-738098	9094	8452	0.929	29.921	9094	1.000	50.101	9094	1.000	49.838
chromaticindex128-5	9217	6914	0.750	31.288	9217	1.000	22.130	9217	1.000	21.805
ta2-UUE	9242	6751	0.730	28.933	9242	1.000	25.406	9242	1.000	21.091
neos-4343293-stony	9401	8769	0.933	29.911	9401	1.000	23.217	9401	1.000	22.104
neos-820879	9523	7971	0.837	33.284	9293	0.976	56.389	9293	0.976	56.800
transportmoment	9686	4952	0.511	35.352	9686	1.000	27.252	9686	1.000	26.767
neos-935769	9800	9661	0.986	35.288	9800	1.000	24.891	9800	1.000	24.925
neos-4647027-thurso	9801	7001	0.714	43.413	9801	1.000	27.415	9801	1.000	26.756
blp-ic97	9846	8358	0.849	39.786	8450	0.858	26.791	8450	0.858	25.751
neos-498623	9862	9715	0.985	36.612	9862	1.000	26.288	9862	1.000	25.467
disctom	10001	9601	0.960	39.619	10001	1.000	63.650	10000	1.000	64.689
n3700	10001	9852	0.985	34.823	10001	1.000	60.145	10001	1.000	61.302
n3705	10001	9852	0.985	36.697	10001	1.000	60.945	10001	1.000	60.555
n3707	10001	9852	0.985	37.371	10001	1.000	61.457	10001	1.000	25.381
n3709	10001	9852	0.985	37.760	10001	1.000	25.542	10001	1.000	59.969
n370b	10001	9852	0.985	36.658	10001	1.000	25.926	10001	1.000	60.360

Table 9 Detailed matrix size reduction across 332 mixed-binary instances